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GENERAL FORMALISM OF WAVE MECHANICS

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We have confined our attention so far to the wave mechanics of a single particle, so that the main features and typical results of the theory can be seen clearly without being clouded by the greater complexity of many-particle systems. We begin this chapter with a brief account of the generalization of the Schrödinger equation to any N-particle system and the interpretation of the wave function in this case. We then proceed to a systematic presentation of the general mathematical framework of wave mechanics and the fundamental postulates (and rules derived therefrom) for the interpretation of the mathematical structures in physical terms. These form the essential basis of quantum mechanics, which any serious student must strive to understand thoroughly.

3.1 THE SCHRÖDINGER EQUATION AND THE PROBABILITY INTERPRETATION FOR AN N-PARTICLE SYSTEM

A system of N particles is characterized by the position and momentum variables $\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_N$ and $\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N$. Its energy is given by the Hamiltonian function H depending in general on all the position and momentum variables:

$$E = H(\mathbf{x}_1, \mathbf{x}_2, ... \mathbf{x}_N; \mathbf{p}_1, \mathbf{p}_2 \ldots \mathbf{p}_N; t) \tag{3.1}$$

The explicit inclusion of the time parameter is intended to take care of the effect of time-dependent external forces acting on the system, if any. Knowing the Hamiltonian, one can write down the Schrödinger equation by a direct generalization of the method used in deriving Eq. (2.19). Firstly, the operator correspondence (2.14) is generalized to

$$E \to i\hbar \frac{\partial}{\partial t}, \qquad \mathbf{p}_i \to -i\hbar \nabla_i \quad (i = 1, 2, ..., N) \tag{3.2}$$

Here ∇_i stands for the gradient operator taken with respect to the position variable $\mathbf{x}_i$ of the ith particle:

$$\nabla_i = \left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial y_i}, \frac{\partial}{\partial z_i} \right)$$

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Since these operators have to act on the wave function ψ, the latter must necessarily depend on all the variables $\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_N}$. This can also be seen from a physical point of view. As we have noted in Sec. 1.17 the wave concept of matter does *not* envisage that the wave functions of different particles add up to a single wave into which they merge completely. The particles remain as separate entities, each with its own set of dynamical variables. This fact must be reflected in the nature of the wave function through a dependence on the position variables of all the particles. It is sometimes convenient to think of the $3N$ coordinates of the N particles as the coordinates of a single point in a $3N$-dimensional space. Such a space is called the *configuration space* of the system. *The wave function ψ is therefore a function defined in the configuration space.*

Using the prescription of Sec. 2.3 we can now write down the equation for the wave function ψ as

$$i\hbar \frac{\partial\psi(\mathbf{x_1}, \mathbf{x_2}, \ldots \mathbf{x_N}; t)}{\partial t}$$

$$= H(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n, -i\hbar\nabla_1, -i\hbar\nabla_2, \ldots, -i\hbar\nabla_N)\psi(\mathbf{x}_1, \ldots, \mathbf{x}_N; t) \tag{3.3}$$

This is the general form of the Schrödinger equation.

It may be noted at this point that though the Hamiltonian can be expressed in terms of generalized coordinates q_α (in the sense of the Lagrangian form of mechanics) and their canonically conjugate momenta p_α. ($\alpha = 1, 2, \ldots, 3N$), the replacement of p_α. by $-i\hbar\partial/\partial q_\alpha$ is correct only if the q_α are *Cartesian* coordinates. This is why we have not used the generalized coordinate notation in Eqs. (3.1)–(3.3). If in a particular problem the use of some generalized coordinates is advantageous (e.g. spherical polar coordinates in systems with spherical symmetry), the operator replacement must be done first with the Hamiltonian expressed in Cartesian coordinates, and only thereafter should the transformation to generalized coordinates be made. A different result would follow if the steps were reversed, and the equation obtained would not be the correct Schrödinger equation.

Example 3.1 The quantum Hamiltonian (operator) for a free particle in three dimensions is $-(\hbar^2/2m)\nabla^2$. When converted to spherical polar coordinates, it takes the form

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right].$$

On the other hand, if we had started with the form of $\mathbf{p}^2$ in spherical polar coordinates, viz.,

$$\mathbf{p}^2 = p_r^2 + \frac{1}{r^2}p_\theta^2 + \frac{1}{r^2\sin^2\theta}p_\varphi^2,$$

and *then* substituted $p_r = -i\hbar(\partial/\partial r)$, $p_\theta = -i\hbar(\partial/\partial\theta)$ and $p_\varphi = -i\hbar(\partial/\partial\varphi)$, we would get a different (and incorrect) expression for the Hamiltonian operator. ■

Example 3.2 What is the Schrödinger equation of a system of two particles of masses m_1 and m_2 carrying charges e_1 and e_2 respectively? The Hamiltonian is obviously $(\mathbf{p}_1^2/2m_1) + (\mathbf{p}_2^2/2m_2) + (e_1e_2/r_{12})$, the last term being the electrostatic interaction energy of the two particles ($r_{12} = |\mathbf{x}_1 - \mathbf{x}_2|$). So the Schrödinger equation is

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$$i\hbar \frac{\partial \psi(\mathbf{x}_1, \mathbf{x}_2; t)}{\partial t} = \left(-\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{\hbar^2}{2m_2} \nabla_2^2 + \frac{e_1 e_2}{r_{12}} \right) \psi(\mathbf{x}_1, \mathbf{x}_2; t)$$

Examples of such systems are: Hydrogen-like atoms (hydrogen, singly ionized helium, doubly ionized lithium, etc.), mesic atoms (a μ^- or π^- meson together with a proton), positronium (electron-positron system). When considering atoms it is usual to consider the position of the nucleus (say $\mathbf{x}_2$) as fixed and the kinetic energy ($\mathbf{p}_2{}^2/2m_2$) to be effectively zero, because of the large nuclear mass (some thousands of times the electron mass). Then the term in $\nabla_2{}^2$ drops out, and the equation reduces to one involving $\mathbf{x}_1$ only, $\mathbf{x}_2$ being just a fixed vector. ■

The many-particle wave function appearing in Eq. (3.3), like that for a single particle, is interpreted in probabilistic terms. Specifically, it is postulated that

$$|\ \psi(\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_N; t)\ |^2 \ d^3x_1 \ d^3x_2 ... d^3x_N \tag{3.4}$$

is the joint probability that particle 1 is within the volume element d^3x_1 around the point $\mathbf{x}_1$, particle 2 is within d^3x_2 around $\mathbf{x}_2$, …, and particle N is within d^3x_N around $\mathbf{x}_N$ at time t. If this expression is integrated over the coordinates of some of the particles, the resulting expression gives the joint probability for finding the remaining particles within specified volume elements irrespective of the positions of the other particles. For example

$$[\int d^3x_2 \ldots \int d^3x_N\, |\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N; t)\ 1^2]\ d^3x_1 \tag{3.5}$$

is the probability that particle 1 is within d^3x_1 when nothing is said about the positions of other particles. If one integrates the expression (3.4) over the coordinates of *all* the particles, one gets the probability that all the particles are to be found somewhere in space, and this should of course be unity:

$$\int d^3x_1 \int d^3x_2 \ldots \int d^3x_N\, |\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathrm{x}_N; t)\ |^2 = 1 \tag{3.6}$$

This *normalization* of the wave function has been assumed in making the statements above regarding the interpretation of ψ. The procedure used in Sec. 2.6 can be generalized in an obvious fashion to prove that the norm is conserved. A simpler proof depending only on a certain general property (self-adjointness) of the Hamiltonian operator will be given in Sec. 3.14 (see the para below Eq. (3.94)). As regards non-normalizable wave functions, remarks similar to those made in Sec. 2.5 apply, and we will not consider them further here.

Associated with the probability interpretation is the postulate regarding the expectation values of dynamical variables. As a direct generalization of Eq. (2.42), we define

$$\langle A \rangle = \iint \ldots \int \psi^*(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathrm{x}_N)\, A_{op}\, \psi\, (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)\, d^3x_1\, d^3x_2 \ldots d^3x_N \tag{3.7}$$

Here A_{op} is obtained by the operator replacement (3.2) from the function $A(\mathbf{x}_1, \ldots, \mathbf{x}_N; \mathbf{p}_1, \ldots, \mathbf{p}_N)$ defining the dynamical variable:

$$A_{op} = A\, (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N; -i\hbar\nabla_1, \ldots, -i\hbar\nabla_N) \tag{3.8}$$

In trying to write down A_{op} explicitly, ambiguities can arise from the fact that the differential operators do not commute with the coordinate variables. As observed in Sec. 2.3, one must then use a symmetrized expression, but we will have no occasion to worry unduly about this problem.

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Notation – To save writing, we will hereafter adopt the following convention: Whenever we are talking of a general system, we will write the wave function $\psi(\mathbf{x_1}, \mathbf{x_2} \dots \mathbf{x_N})$ in abbreviated form as $\psi(\mathbf{x})$, it being understood that $\mathbf{x}$ stands for the coordinates $\mathbf{x_1}, \mathbf{x_2} \dots \mathbf{x_N}$ of all the particles. Similarly, we will write $\mathbf{p}$ for the set of momentum variables $\mathbf{p_1}, \mathbf{p_2}, \dots, \mathbf{p_N}$. The individual volume elements d^3x_1, d^3x_2 … will be denoted by $d\tau_1, d\tau_2 \dots$, and we will use $d\tau$ as an abbreviation for $d\tau_1, d\tau_2 \dots d\tau_N$. Thus Eqs (3.6) and (3.7) would appear in this shorthand notation as

$$\int d\tau\, \psi^*(\mathbf{x})\psi(\mathbf{x}) = 1, \qquad \langle A \rangle = \int d\tau\, \psi^*(\mathbf{x})\, A_{op}\, \psi\,(\mathbf{x}) \tag{3.9}$$

This convention will help us to avoid using cumbersome expressions in discussing general theorems. Of course in particular problems where the number of particles involved is important, the expressions will be written out in full.

3.2 THE FUNDAMENTAL POSTULATES OF WAVE MECHANICS

What we have learned so far about the nature of the wave-mechanical description of physical systems and about the rules of interpretation, were deduced from observational facts by heuristic (non-rigorous) arguments. Beginning with a formal statement of these as basic postulates, we shall now study their consequences in some detail. In the course of this study, the mathematical structure of wave mechanics and its physical content will be made more explicit.

(*a*) *Representation of states*

Postulate 1 – The *state* of a quantum mechanical system *is described or represented by a wave function* $\psi\,(\mathbf{x}, t)$.

The wave function contains all the information which nature permits one to give about the state of the system at any instant of time. We will often speak of the 'state ψ' meaning thereby the physical concept (the state) described by the mathematical function $\psi(\mathbf{x}, t)$. The wave function is not directly measurable. In fact, all *constant* multiples of a given ψ (namely $c\psi$, for all real or complex numbers c) describe one and the same state.

Postulate 1′ – The Superposition Principle. If ψ_1 and ψ_2 are wave functions describing any two states of a given system, then corresponding to every linear combination[1] $(c_1\psi_1 + c_2\psi_2)$ of the two functions there exists a state of the system.

This is a fundamental principle of quantum mechanics to which there is no correspondence in classical mechanics. It is the possibility of superposition which makes interference phenomena possible. We have already assumed it in constructing wave packet states of a free particle, as in Eq. (1.31), as superpositions of de Broglie waves corresponding to free particle states of various momenta.

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[1] The term 'linear combination' will appear repeatedly in the text. By a linear combination of $\psi_1, \psi_2, \dots, \psi_r$ we mean any quantity of the form $c_1\psi_1 + c_2\psi_2 + \dots c_r\psi_r$ where $c_1, c_2, \dots, c_r$ are any complex numbers. We say that $\psi_1, \psi_2, \dots, \psi_r$ form a *linearly independent set* if none of them can be expressed as a linear combination of the others; or equivalently, if the only way to make $c_1\psi_1 + c_2\psi_2 + \dots c_r\psi_r$ vanish is by taking *all* the constant coefficients to be zero.

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The principle of superposition has deep mathematical significance too. This realization comes from the observation that this principle is a fundamental characteristic of *vectors* representing points in space. (If $\mathbf{v}_1, \mathbf{v}_2$ are the position vectors of any two points, $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ is again a vector which represents some point in space). Therefore, it is possible (and useful) to think of wave *functions* as vectors of some kind. The ordinary vectors which we are familiar with need only three real numbers each, for a complete specification; the components v_1, v_2, v_3 identify a vector $\mathbf{v}$ completely. But each wave function-vector requires an *infinite* set of *complex* numbers for its specification, namely the values of ψ *at all the points* $\mathbf{x}$ in (configuration) space. Therefore, we say that the function-vectors are *infinite-dimensional.* The set of all wave functions of a given system (which includes also all linear combinations of the wave functions) forms an infinite dimensional *complex linear vector space.*

In this 'geometrical' picture of wave functions, the quantities $\psi(\mathbf{x})$ for various $\mathbf{x}$ are thought of as different 'components' of the wave function-vector ψ. Here the continuous variable $\mathbf{x}$ serves as a label distinguishing the different components of ψ, and is the counterpart of the index i labelling the components of v_i an ordinary vector $\mathbf{v}$. With this identification, the norm $\int |\psi(\mathbf{x})|^2 d\tau$ becomes the analogue of $\Sigma_i v_i^2$, and it may therefore be viewed as the 'square of the length' of the function-vector. (Since $\psi(\mathbf{x})$ is complex, the absolute square $|\psi(\mathbf{x})|^2$ has to be taken to get a real 'length' and of course, it is 'summation' over $\mathbf{x}$ which is indicated by the integral sign). More generally, $\int \phi^*(\mathbf{x})\, \psi(\mathbf{x})\, d\tau$ is defined as the *scalar product* of ϕ and ψ, analogous to $\Sigma_i u_i v_i$ for ordinary vectors. In view of this, the notation (ϕ,ψ)– as the counterpart of $\mathbf{u}.\mathbf{v}$– is often used for such integrals. From the definition

$$(\phi, \psi) \equiv \int \phi^*(\mathbf{x})\psi(\mathbf{x})\, d\tau \qquad \text{(scalar product of } \phi, \psi),$$

it follows that

$$(\phi, \psi) = (\psi, \phi)^*,$$

$$(\phi, c\psi) = c(\phi, \psi), \quad (c\phi, \psi) = c^*(\phi, \psi),$$

and that

$$\text{Norm of } \psi = (\psi, \psi) \geqslant 0$$

with the equality sign holding if and only if $\psi \equiv 0$.

The above notation is now in common usage. To help the student in gaining familiarity with it, we shall frequently write the scalar product notation (ϕ, ψ) side by side with its explicit integral expression.

(*b*) *Representation of dynamical variables; Expectation values, Observables*

Postulate 2 – Each *dynamical variable* $A(\mathbf{x}; \mathbf{p})$ is represented in quantum mechanics by a *linear operator*[2] $A_{op} = A(\mathbf{x}_{\text{op}}, \mathbf{p}_{\text{op}}) \equiv A(\mathbf{x}, -i\hbar\nabla)$.

[2] We will drop the subscript 'op' hereafter, and write, for example, $\mathbf{p} = -i\hbar\nabla$ which is to be understood as '$\mathbf{p}$ is represented by the operator $\mathbf{p}_{op} = -i\hbar\nabla$'. Therefore, the same symbol will hereafter be used for both the physical concept of the dynamical variable (as when we say 'the momentum $\mathbf{p}$') and for the quantum operator representing it (as when we write $\mathbf{p}\psi$ meaning $-i\hbar\nabla\psi$). Which of these is meant in a particular case will be clear from the context. " There exist also dynamical variables that are not functions of $\mathbf{x}_{op}$, $\mathbf{p}_{op}$; they will be considered below in subsection (*d*)."

The operators act on the wave functions of the system. The effect of an operator A on a wave function ψ is to convert it into another wave function, denoted by $A\psi$. *Linearity* of the operator means that a linear combination of two (or more) wave functions, say ψ_1 and ψ_2 is converted into the *same* linear combination of $A\psi_1$ and $A\psi_2$:

$$A\,(c_1\psi_1 + c_2\psi_2) = c_1\,(A\psi_1) + c_2\,(A\psi_2) \tag{3.10}$$

Dynamical variables in quantum mechanics do not, in general, commute with each other: $AB \neq BA$. By this we mean that $A_{op}B_{op}\psi \neq B_{op}A_{op}\psi$ for *arbitrary* wave functions ψ (though there may be particular functions for which the two sides are equal). The difference $AB - BA$ is called the *commutator* of A and B. A bracket notation is used to denote the commutator

$$[A, B] \equiv AB - BA \tag{3.11}$$

Commutation relations (relations giving the values of the commutators) of position and momentum variables may be easily deduced. For example, in one dimension,

$$(xp - px)\,\psi = -\,i\hbar\left[x\frac{\partial}{\partial x}\psi - \frac{\partial}{\partial x}(x\psi)\right] = i\hbar\psi$$

Thus the operator· $[x, p]$ has the effect of simply multiplying any arbitrary wave function ψ by $i\hbar$ and we can therefore say that $[x, p] = i\hbar$. In three dimensions, if x_i and $p_j \equiv -\,i\hbar\,\partial/\partial x_j$ are the respective operators representing the ith and jth components of the position and momentum of a particle,[3] we have

$$x_i p_j \psi = -i\hbar\; x_i\,\frac{\partial\psi}{\partial x_j},$$

and

$$p_j x_i \psi = -i\hbar\,\frac{\partial}{\partial x_j}(x_i\psi) = -i\hbar\left[\delta_{ij}\psi + x_i\frac{\partial\psi}{\partial x_j}\right],$$

since $\partial x_i/\partial x_j$ is equal to the *Kronecker delta function* δ_{ij} *defined as:*

$$\delta_{ij} = 1 \quad \text{if } i = j, \quad \text{and} \quad \delta_{ij} = 0 \quad \text{if } i \neq j \tag{3.12}$$

Thus $(x_ip_j - p_jx_i)\,\psi = i\hbar\,\delta_{ij}\psi$. Since this is true for *any* wave function ψ we can conclude that $x_ip_j - p_jx_i = i\hbar\delta_{ij}$ or

$$[x_i, p_j] = i\hbar\,\delta_{ij} \tag{3.13a}$$

It is a trivial matter to verify in a similar manner that

$$[x_i, x_j] = 0, \quad [p_i, p_j] = 0 \tag{3.13b}$$

[3] We will use either of the notations (x, y, z) or (x_1, x_2, x_3) for the components of $\mathbf{x}$, as convenience dictates. Similarly the components of $\mathbf{p}$ will be written as (p_x, p_y, p_z) or (p_1, p_2, p_3).

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The above equations assert that of all the commutators among the position and momentum components *the only non-vanishing ones* are

$$[x, p_x] = [y, p_y] = [z, p_z] = i\hbar \tag{3.13c}$$

These equations may be considered as definitions of *canonically conjugate pairs* of dynamical variables in quantum mechanics. More generally, if subscripts α, β denote the αth and βth particles of a many particle system, we have

$$\begin{aligned}[x_{\alpha i}, p_{\beta j}] &= i\hbar\delta_{\alpha\beta}\delta_{ij};\\ [x_{\alpha i}, x_{\beta j}] &= [p_{\alpha i}, p_{\beta j}] = 0\end{aligned} \tag{3.14}$$

These commutation relations are fundamental to all of quantum mechanics. Commutators involving products of components of position coordinates and momenta can be evaluated using these basic commutation relations and the identities

$$\begin{aligned}[AB, C] &= A[B, C] + [A, C]B\\ [A, BC] &= [A, B]C + B[A, C]\end{aligned} \tag{3.15}$$

which are valid for arbitrary operators A, B, C.

Example 3.3 Let us evaluate $[x, p^n]$ in the one-dimensional case from the basic relation $[x, p] = i\hbar$ using Eqs (3.15). We have $[x, p^2] = p[x, p] + [x, p]p = 2i\hbar p$ and then $[x, p^3] = [x, p^2p] = p^2[x, p] + [x, p^2]p = 3i\hbar p^2$. Continuing this process, we get $[x, p^n] = i\hbar\, n\, p^{n-1}$. By a similar process we can show that $[p, x^n] = -i\hbar n x^{n-1}$. Actually this result is more easily obtained by direct use of the operator form $(-i\hbar d/dx)$ of p, as in the derivation of the commutation rules (3.13). The student may verify that for any function $f(x)$, $[p, f(x)] = -i\hbar df(x)/dx$. In three dimensions,

$$[\mathbf{p}, f(\mathbf{x})] = -i\hbar\nabla f(\mathbf{x})$$

and similarly,

$$[\mathbf{x}, f(\mathbf{p})] = i\hbar\nabla_p f(\mathbf{p}),$$

where ∇p stands for $(\partial/\partial p_x, \partial/\partial p_y, \partial/\partial p_z)$. ■

Example 3.4 In taking the square (or higher powers) of a product of operators, possible noncommutativity of the factors should not be overlooked. For example, $(xp_x)^2 \neq x^2p_x^2$. Actually $(xp_x)^2 = xp_xxp_x = x(xp_x - i\hbar)\,p_x = x^2p_x^2 - i\hbar xp_x$.

Similarly

$$(xp_x)\,(yp_y) = xp_x\,(p_yy + i\hbar) = (xp_y)\,(yp_x) + i\hbar xp_x.$$

Using results of this kind we can express the expansion of $(\mathbf{x.p})^2$ as

$$\begin{aligned}(\mathbf{x.p})^2 = (x^2 + y^2 + z^2)\,(p_x^2 + p_y^2 + p_z^2) - (xp_y - yp_x)^2\\ - (yp_z - zp_y)^2 - (zp_x - xp_z)^2 + i\hbar\,(xp_x + yp_y + zp_z)\end{aligned}$$

or

$$(\mathbf{x.p})^2 = \mathbf{x^2p^2} - \mathbf{L}^2 + i\hbar\,(\mathbf{x.p}).$$

This result will be used in Sec. 4.6. ■

(*c*) *Angular momentum operators; Commutation relations:* The components of the angular momentum vector associated with the motion of a particle in space are dynamical variables of fundamental importance. The theory of angular momentum is an essential part of the machinery for solving problems relating to particles moving in spherically symmetric potentials. The ground work for application of theory to such problems is laid in Part B of Chapter 4 of this book. Application to specific problems of great interest is done in subsequent sections of that chapter. Chapter 8 presents the theory of angular momentum in all generality.

We go on now to the evaluation of the commutators involving components of the angular momentum vector $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, namely,

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x \tag{3.16}$$

Let us consider, for instance, the commutators of L_z with the components x, y, z of the position vector $\mathbf{x}$. By applying the general commutation rules (3.15), we obtain

$$\begin{aligned}[L_z, x] &= [(xp_y - yp_x), x] = [x, x]p_y + x[p_y, x] - [y, x]p_x - y[p_x, x] \\ &= -y[p_x, x] = i\hbar y\end{aligned} \tag{3.17}$$

Proceeding in a similar fashion, we can readily evaluate the commutators of L_z with y and z as well as with the components of $\mathbf{p}$. We find thus that

$$[L_z, x] = i\hbar y, \qquad [L_z, y] = -i\hbar x, \qquad [L_z, z] = 0 \tag{3.18}$$

$$[L_z, p_x] = i\hbar p_y, \qquad [L_z, p_y] = -i\hbar p_x, \qquad [L_z, p_z] = 0 \tag{3.19}$$

The commutators of L_x and L_y with the components of $\mathbf{x}$ and $\mathbf{p}$ can be obtained simply by cyclic permutations $(x, y, z) \to (y, z, x) \to (z, x, y)$ of the symbols (x, y, z) which appear in the above pair of equations, whether as subscripts or as variables. For example, the first of the above permuations takes Eq. (3.18) to

$$[L_x, y] = i\hbar z, \qquad [L_x, z] = -i\hbar y, \qquad [L_x, x] = 0 \tag{3.20}$$

The student may carry on with the application of the next permutation to this set of equations, and then write down explicitly also the parallel for the commutators with p_x, p_y, p_z.

The complete set of commutation relations, thus obtained, may be employed directly to obtain the commutators of the angular momentum components among themselves. For example, we find by using the analogues of Eqs (3.18) and (3.19) for the commutators of L_x and applying the commutation rules (3.15) that

$$[L_x, L_y] = [L_x, (zp_x - xp_z)] = [L_x, z]p_x - x[L_x, p_z] = i\hbar(-yp_x + xp_y) = i\hbar L_z \tag{3.21}$$

The commutators of the other pairs of components may be evaluated similarly. We finally have

$$[L_x, L_y] = i\hbar L_x, \qquad [L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{3.22}$$

These are the fundamental commutation relations of the angular momentum operators. Note that we have not had to use the differential operator form for $\mathbf{p}$ to obtain any of the above results; in fact it is usually much simpler to use the commuation rules given in Eqs (3.13) directly than to go back to $\mathbf{p} = -i\hbar\nabla$.

The above commutation relations hold also for the total angular momentum of $\mathbf{L} = \sum_\alpha \mathbf{L}_\alpha$ of a system of particles. The student is urged to verify this, keeping in mind that the angular momenta $\mathbf{L}_\alpha$ and $\mathbf{L}_\beta$ of two different particles ($\alpha \neq \beta$) commute in view of (3.14).

Example 3.5 When written explicitly as a differential operator, L_z can be written as equal to $-\ i\hbar\ (x\partial/\partial y - y\partial/\partial x)$. On transforming this operator to spherical polar coordinates, we get $L_z = —i\hbar\partial/\partial\varphi$. Hence we can formally write $[\varphi, L_z] = i\hbar$. This relation has the same form as Eq. (3.13c). Nevertheless φ and L_z should not be considered as canonically conjugate variables; indeed φ *itself cannot be regarded as a proper dynanmical variable* in quantum mechanics. The reason is that it has the effect of converting a perfectly good wave function ψ into one with inadmissible properties. For, if φ is limited to the interval 0 to 2π, then after it has increased to 2π it jumps discontinuously down to zero, and the function $\varphi\psi$ will do the same. On the other hand, if φ is allowed an unlimited range of variation, $\varphi\psi(r, \theta, \varphi)$ will violate the single-valuedness condition. Thus $\varphi\psi$ is not an admissible wave function though ψ is, which is to say that φ is not a good operator. As a corollary, one cannot draw the same kind of conclusions from the formal relation $[\varphi, L_z] = i\hbar$ as can be done from Eq. (13.13c). In particular it is not possible to infer an uncertainty relation $(\Delta\varphi)\ (\Delta L_z) \geqslant \frac{1}{2}\hbar$ analogous to Eq. (3.80). ■

(*d*) *More general operators*: There are, in quantum mechanics, certain dynamical variables (e.g. parity) which do not have a classical counterpart, i.e. there is no classical function $A(\mathbf{x}, \mathbf{p})$ which would yield the quantum operator for such a quantity through the operator replacement of postulate 2. In these cases the operators concerned are defined directly by stating what effect they produce on arbitrary states or wave functions of the system. (See, for example, Sec. 4.11 for parity). The commutation rules (3.13) have nothing to say about commutation relations involving such operators, which must be deduced directly from their definitions. We will consider these explicitly at the appropriate stage, but it may be kept in mind that the treatment of the remainder of this chapter holds good also for such nonclassical dynamical variables.

Finally, we remind the reader of the observation at the end of Sec. 1.11 that electrons (and other particles like the proton and neutron) carry *spin* angular momentum, i.e., angular momentum associated with what may be visualized as a spinning motion of the particle analogous to the spinning of a top or the rotation of the earth about its own axis. This spinning motion is of course quite independent of the position and momentum of linear motion of the particle. Therefore one needs new dynamical variables (which *cannot* be functions of $\mathbf{x}$ and $\mathbf{p}$) to represent this degree of freedom. Further, the concept of the wave function itself has to be generalized to include a dependence on the direction of spin also (besides the dependence on $\mathbf{x}$). Fortunately, in a wide variety of important phenomena, the presence of spin plays no significant role. So we postpone consideration of this complication to Sec. 8.3. However the general theorems of this chapter hold good for the spin operators too.

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POSTULATE 3– If a large number of measurements of a dynamical variable A are made on a system which is prepared to be in one and the same state ψ before each measurement, the results of the individual measurements will in general be different, but the average of all the observed values is expected to be

$$\langle A\rangle_\psi = \int \psi^* A\psi d\tau \equiv (\psi, A\psi) \tag{3.23a}$$

The quantity $\langle A\rangle_\psi$ is called the *expectation value* of the dynamical variable A. Note that inside the integral, A means A_{op}.

On inspection of the special case of Eq. (3.23a) when A is the position variable of any of the particles, it becomes evident that the definition of the expectation value automatically implies the interpretation of $|\psi|^2$ as the probability density in configuration space, and the normalization of ψ according to Eq. (3.6). Unless otherwise stated, all wave functions will be taken to be so normalized. For the present we will assume that non-normalizable wave functions are also brought into this category through suitable limiting procedures like the technique of box normalization (Sec. 2.5). Later in this chapter we will consider the kind of normalization involving the Dirac delta function, which becomes necessary if unnormalizable functions are to be handled directly. Whenever a ψ having a different normalization from (3.6) is employed for any reason, the definition (3.23a) of expectation values is to be replaced by

$$\langle A\rangle = \frac{\int \psi^* A\psi d\tau}{\int \psi^* \psi d\tau} \tag{3.23b}$$

The expectation value $\langle A\rangle$ may be real or complex, depending on the nature of A. However, from the experimental point of view, the result of any single observation or the measurement of a single quantity is necessarily a real number (e.g. the reading on a dial or scale, weights in a pan, etc). To measure any complex-valued variable, one has to separate it into two real variables (the real and the imaginary parts, or the modulus and the argument), and measure each separately. The measurement of each part always gives some real number as the result; and the average result of repeated measurements is also of course real. This fact implies, in the quantum mechanical context, that only those dynamical variables whose expectation values are real should be considered to be directly measurable, or *observable.* Thus if A is to be observable, it is necessary that $\langle A\rangle = \langle A\rangle^*$, or

$$\int \psi^* A\psi d\tau = \int (\psi^* A\psi)^* d\tau = \int (A\psi)^* \psi d\tau \tag{3.24}$$

If this is to be ensured for all wave functions ψ, the operator A must have a certain mathematical property called self-adjointness, which will be defined in the next section.

The three postulates considered so far concern the kinematics (description at a particular time) of a quantum system. The fourth basic postulate concerns the dynamics (development with time). This will be taken up in Sec. 3.14, after the discussion of kinematical questions is completed.

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Example 3.6 To evaluate $\langle x^n \rangle$ when ψ is the Gaussian function, namely $\psi = (1/\sigma\sqrt{\pi})^{1/2}$. $\exp[-x^2/2\sigma^2]$:

$$(\sigma\sqrt{\pi})\left\langle x^n \right\rangle = \int_{-\infty}^{\infty} x^n e^{-x^2/\sigma^2} dx$$

$$= \int_{0}^{\infty} x^n e^{-x^2/\sigma^2} dx + \int_{-\infty}^{0} x^n e^{-x^2/\sigma^2} dx$$

$$= [1 + (-1)^n] \int_{0}^{\infty} x^n e^{-x^2/\sigma^2} dx .$$

The last expression has been obtained with the aid of the substitution $x = -x$ in the second integral in the previous step. We thus have immediately that

$$\langle x^n \rangle = 0 \text{ for odd values of } n.$$

This result is a consequence of the fact that the integrand, $x^n \exp[-x^2/\sigma^2]$ is of odd parity for odd n. It is true quite generally that the integral (from $-\infty$ to $+\infty$) of any function of odd parity is zero. This is a useful fact to remember for future calculations. From this fact we can directly conclude, for instance, that $\langle p \rangle = 0$ since in $\langle p \rangle = \int \psi^*(-i\hbar d/dx)\psi \, dx$, the Gaussian ψ is of even parity while $d\psi/dx$ is of odd parity.

To determine $\langle x^n \rangle$ when n is an even integer, say $n = 2s$, evaluation of the final integral is best done by converting it to the form of a gamma function, $\Gamma(m) = \int_0^\infty e^{-w} w^{m-1} dw$, by the substitution $(x^2/\sigma^2) = w$. Remembering $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(m) = (m - 1)\,\Gamma(m - 1)$ we get

$$\langle x^{2s} \rangle = \frac{(2s)!}{s!} \cdot \left(\frac{\sigma}{2}\right)^{2s}, \qquad (s = 0, 1, 2, \ldots).$$

Note that the value unity of $\langle x^n \rangle$ for $n = 0$ checks that the wave function is correctly normalized. ■

3.3 THE ADJOINT OF AN OPERATOR AND SELF-ADJOINTNESS

Let us consider the integral

$$\int \phi^* A \psi d\tau \equiv (\phi, A\psi)$$

which involves two different functions ϕ and ψ and reduces to (3.23) in the special case when $\phi = \psi$. It can be shown that one can always find another operator, called the *adjoint* of A and denoted by $A^\dagger$ (read A dagger), such that

$$\int \phi^* A \psi \, d\tau = \int (A^\dagger \phi)^* \psi \, d\tau, \quad \text{or} \quad (\phi, A\psi) = (A^\dagger \phi, \psi) \tag{3.25}$$

In other words, as far as the value of the integral is concerned, it makes no difference whether A acts on ψ or its adjoint $A^\dagger$ acts on the other wave function ϕ. Equation (3.25) serves to define the adjoint of any operator. From this definition it is easy to show

$$(A + B)^\dagger = A^\dagger + B^\dagger \tag{3.26}$$

and that if c is a complex number

$$(cA)^\dagger = c^* A^\dagger \tag{3.27}$$

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i.e. in taking the adjoint, any complex number goes over into its complex conjugate. Further, since

$$\int\phi^*(A^\dagger\psi)\,d\tau = \left[\int(A^\dagger\psi)^*\phi\,d\tau\right]^* = \left[\int\psi^*A\phi\,d\tau\right]^* = \int(A\phi)^*\psi\,d\tau$$

we have

$$(A^\dagger)^\dagger = A \tag{3.28}$$

For the adjoint of the product of two operators A and B, by applying the definition given in Eq. (3.25) successively to A and B, one obtains

$$\int\phi^*AB\psi d\tau = \int(A^\dagger\phi)^*\,B\psi d\tau = \int(B^\dagger A^\dagger\phi)^*\psi d\tau,$$

or

$$(AB)^\dagger = B^\dagger A^\dagger \tag{3.29}$$

An operator A is said to be *self adjoint*[4] if its adjoint is equal to itself:

$$A^\dagger = A,$$

or

$$\int\phi^*A\psi\,d\tau = \int(A\phi)^*\psi\,d\tau, \quad \text{i.e., } (\phi, A\psi) = (A\phi, \psi) \tag{3.30}$$

Note that the product of two self-adjoint operators is *not* necessarily self-adjoint. For, if $A^\dagger = A$ and $B^\dagger = B$, then according to Eq. (3.27),

$$(AB)^\dagger = BA \tag{3.31}$$

Thus AB is self adjoint only if $BA = AB$, i.e., if A and B commute. However, the two combinations

$$(AB + BA) \quad \text{and} \quad i(AB - BA) \tag{3.32}$$

are both self-adjoint, as the reader may verify.

It is now a trivial matter to see that the expectation value of a self-adjoint operator is real, for on setting $\phi, = \psi$ in Eq. (3.30) it reduces to $\langle A\rangle = \langle A\rangle^*$. Thus self-adjoint operators are suitable for representing observable dynamical variables, and we proceed to study their most important properties. Incidentally, it may be observed that $A^\dagger A$ is always self-adjoint (even if A is not). Further, its expectation value is nonnegative in *all* states. (Any operator with this property is said to be *positive*.)

$$\langle A^\dagger A\rangle = \int\psi^*A^\dagger A\,\psi\,d\tau = \int(A\psi)^*\,(A\psi)\,d\tau \geqslant 0 \tag{3.33a}$$

since the integrand, heing the absolute square of the function $A\psi$, is nonnegative. Evidently if $\langle A^\dagger A\rangle$ is to vanish, the integrand must vanish identically. Thus,

$$\langle A^\dagger A\rangle = 0 \text{ implies } A\psi = 0 \tag{3.33b}$$

[4] For reasons which will become clear later, in the context of quantum mechanics, the term *hermitian* operator is very often used for self-adjoint operators. We will also be using the words 'hermiticity' and 'self-adjointness' interchangeably.

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Example 3.7 The position operator is obviously self-adjoint. As for the momentum operator, we find by partial integration that

$$\int \phi^* p\psi\, dx \equiv \int \phi^*(-i\hbar)\frac{d\psi}{dx}\, dx$$
$$= -i\hbar\phi^*\psi + \int \left(-i\hbar\frac{d\phi}{dx}\right)^* \psi\, dx$$

The first term drops out in the case of normalizable wave functions ϕ, and ψ since they vanish at both the limits of integration ($x \to \pm\infty$). Then the expression reduces to $\int (p\phi)^*\psi dx$ showing that p is self-adjoint. In the case of box-normalized functions, the condition that all wave functions be periodic within the box (see Sec. 2.5) ensures that ($\phi^*\psi$) cancels out between the upper and lower limits ($x = —L/2$ and $+ L/2$). Therefore **p** is self-adjoint in this case too.

Knowing that **x** and **p** are self-adjoint, one can easily prove the same property for **L** and H (provided the potential function in the latter is a *real* function of **x**). ■

3.4 THE EIGENVALUE PROBLEM; DEGENERACY

Further development of the formalism relies heavily on the concept of eigenfunctions and eigenvalues. We had occasion to consider these in the context of the energy or Hamiltonian operator in Sec. 2.9. In general, for any operator A one can, set up the eigenvalue equation

$$A\phi_a = a\phi_a \tag{3.34}$$

Let us recall the relevant definitions. If a function ϕ_a is such that the action of the operator A on it has the simple effect of multiplying it by a constant factor a, then we say that ϕ_a is an *eigenfunction of A belonging to the eigenvalue a.* The set of all eigenvalues of A form what is called the *eigenvalue spectrum* (or simply the spectrum) of A. The spectrum may be continuous, or discrete, or partly continuous and partly discrete (as is the case with the energy spectrum for the square well potential). If there exists only one eigenfunction belonging to a given eigenvalue, the eigenvalue is said to be *non-degenerate;* otherwise it is *degenerate.*

Associated with any degenerate eigenvalue there is always an *infinite* number of eigenfunctions. For example, if ϕ_a, χ_a belong to the same eigenvalue a, then so do the combinations $c_1\phi_a + c_2\chi_a$ for *all* values of c_1 and c_2:

$$A(c_1\phi_a + c_2\chi_a) = c_1A\phi_a + c_2A\chi_a = a\,(c_1\phi_a + c_2\chi_a)$$

Since the set of all eigenfunctions belonging to a given degenerate eigenvalue a is *closed under linear combinations* (i.e. any linear combination of members of the set is again a member), this set forms a linear space. This space may be called the eigenspace belonging to the eigenvalue a of A. We can always choose from this function-space a linearly independent subset of functions, say $\phi_{a_1}, \phi_{a_2}..., \phi_{a_r}$, such that *every* eigenfunction belonging to a can be expressed uniquely as a linear combination

$$c_1\phi_{a_1} + c_2\phi_{a_2} + \ldots + c_r\phi_{a_r}$$

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with suitable coefficients c_1, c_2,...c_r. We say then that ϕ_{a_1}, ϕ_{a_2}, ... ϕ_{a_r} form a set of basis functions (or simply a *basis*) which *spans* the linear space. There is an infinite number of ways of choosing a basis, just as there is an infinite number of ways of choosing coordinate axes in three-dimensional space. But the number r is characteristic of the space. This means, in the present context, that among the infinite number of eigenfunctions belonging to a given degenerate eigen value, there is a definite number r of linearly independent ones. This number is called the *degree of degeneracy* of the eigenvalue; we say that the eigenvalue is *r-fold degenerate.*

The task of determining all the eigenvalues and eigenfunctions of any operator is referred to as the eigenvalue problem. We will be solving a number of important eigenvalue problems in the next chapter. Here we proceed with the consideration of general properties of eigenvalues and eigenfunctions.

Example 3.8 To see the significance of eigenvalues and eigenfunctions, consider an eigenstate ϕ_a of A. Then $A^n\phi_a = a^n\phi_a$, and hence $\langle A^n\rangle = a^n = \langle A\rangle^n$, i.e. in this state, the average value of the nth power of A is just the nth power of $\langle A\rangle$. This implies that when the system is in an *eigenstate* of A, the dynamical variable A has a *definite value equal to the eigenvalue* to which the state belongs. This result will be confirmed in a different way later. ■

Example 3.9 In one dimension, the eigenvalue equation for the energy of a free particle is $(-\hbar^2/2m)\,(d^2\phi/dx^2) = E\phi$, or $d^2\phi/dx^2 + \alpha^2\phi = 0$, with $\alpha^2 = (2mE/\hbar^2)$. For any given E, this equation has *two independent* solutions, $e^{i\alpha x}$ and $e^{-i\alpha x}$. So each eigenvalue is *doubly degenerate.* Of course all linear combinations of these are also solutions, e.g. $\sin\alpha x$, $\cos\alpha x$. The set $e^{i\alpha x}$, $e^{-i\alpha x}$ forms a basis for the space of these solutions. So also does $\sin\alpha x$ and $\cos\alpha x$ or $e^{i\alpha x}$ and $\cos\alpha x$, etc. ■

3.5 EIGENVALUES AND EIGENFUNCTIONS OF SELF-ADJOINT OPERATORS

Let A be a self-adjoint operator, and ϕ_a, ϕ_a' be two of its eigenfunctions:

$$A\phi_a = a\phi_a,\ A\phi_a' = a'\phi_a' \tag{3.35}$$

Substituting $\phi = \phi_a$ and $\psi = \phi_a'$ in the self-adjointness condition (3.30), we get

$$\int \phi_a^*\, A\phi_a'\, d\tau; = \int (A\phi_a)^*\phi_a'\, d\tau \tag{3.36}$$

Using Eq. (3.35) we replace the operator A in the integral by the relevant eigenvalues. Equation (3.36) then reduces to

$$(a' - a^*) \int \phi_a^*\phi_a'\, d\tau = 0 \tag{3.37}$$

This is valid for any two eigenfunctions of A. Let us consider the special case $\phi_a' = \phi_a$, which automatically implies $a' = a$. Then

$$(a - a^*) \int \phi_a^*\, \phi_a\, d\tau = 0 \tag{3.38}$$

The integral here is the norm of ϕ_a which can never vanish. Therefore,

$$a = a^* \tag{3.39}$$

Thus the eigenvalues of a self-adjoint operator are real.

Feeding this information back into Eq. (3.37) we conclude that

$$\int \phi_a{}^* \phi_a{}' d\tau = 0 \quad \text{or} \quad (\phi_a, \phi_a{}') = 0 \quad \text{for} \quad a \neq a' \tag{3.40}$$

This result may be stated in words as follows:

Any two eigenfunctions belonging to distinct (unequal) eigenvalues of a self-adjoint operator are mutually orthogonal[5].

If a and a' have the same value but ϕ_a is different from $\phi_a{}'$, (i.e. if ϕ_a and $\phi_a{}'$, belong to the same degenerate eigenvalue), we can draw no conclusions from Eq. (3.38). In fact, it is not to be expected that all the (infinite number of) eigenfunctions belonging to a degenerate eigenvalue will be mutually orthogonal. However, *it is always possible to choose* from among these *a linearly independent set of basis functions which are mutually orthogonal* (see Example 3.10 below).

In order to keep the essentials of the further development clear, in the discussions of the next few sections, we will assume that all eigenvalues of A are non-degenerate. The elaborations which are necessary when degenerate eigenvalues are present will be carried out in Sec. 3.13.

So far, we have assumed nothing about the norm of the eigenfunctions. Actually we know from the example of the square well potential problem considered in Chapter 2 that an operator may have both normalizable and non-normalizable eigenfunctions. Thus the norm of ϕ_a here may be either 1 or ∞. We, therefore, write

$$\int \phi_a{}^* \phi_a{}' d\tau = \delta(a, a') \tag{3.41a}$$

where $\delta(a, a') = 0$ for $a \neq a'$ and $\delta(a, a) = 1$ or ∞ according as ϕ_a is, or is not, normalizable. In the former case $\delta(a, a')$ is clearly nothing but the Kronecker delta function:

$$\delta(a, a') = \delta_{aa'} \tag{3.41b}$$

In the latter case (eigenfunctions of infinite norm) we write

$$\delta(a, a') = \delta(a - a') \tag{3.41c}$$

where $\delta(a - a')$ is the Dirac delta function, defined below. We will see shortly that Eq. (3.41b) applies if a belongs to the discrete part of the eigenvalue spectrum, and (3.41c) in the case of eigenvalues belonging to the continuum.

Example 3.10 From an arbitrary linearly independent set of normalized functions $\chi_1, \chi_2, \ldots, \chi_r$ we can construct a new set of *orthogonal* functions $\psi_1, \psi_2, \ldots, \psi_r$ by the *Schmidt Orthogonalization Procedure.* Understanding of this procedure is facilitated by recalling that normalized wave functions can be thought of as unit vectors in a function space. To construct from an ordinary vector **V** another vector orthogonal to a given *unit* vector **U**, all we have to do

[5] The term 'orthogonal' is taken from the language of vector spaces; it means that the scalar product (of the function-vectors $\phi_a, \phi_a{}'$, in the present case) vanishes.

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is to remove from **V** its component parallel to **U**, namely **U** (**U**.**V**). The result is **V** —**U** (**U**.**V**). In analogy with this we have on taking $\psi_1 \equiv \chi_1$ in place of U and χ_2 instead of **V**,

$$\psi_2' = \chi_2 - \psi_1 (\psi_1,\chi_2) \equiv \chi_2 - \psi_1 \int \psi_1^* \chi_2 \, d\tau$$

It is a trivial matter to satisfy oneself that ψ_2' is indeed orthogonal to ψ_1, i.e. $(\psi_1, \psi_2') = 0$. Normalization of ψ_2' may be carried out as usual: we have

$$\psi_2 = \psi_2' \Big/ \left[\int \psi_2'^* \psi_2' \, d\tau\right]^{1/2} \equiv \psi_2'/(\psi_2', \psi_2')^{1/2}$$

This procedure may be continued by subtracting from χ_3 the parts 'parallel' to ψ_1 and ψ_2 and then normalizing:

$$\psi_3' = \chi_3 - \psi_1(\psi_1, \psi_3) - \psi_2(\psi_2, \psi_3),$$

$$\psi_3 = \psi_3'/(\psi_3', \psi_3')^{1/2},$$

and so on. ■

3.6 THE DIRAC DELTA FUNCTION

The Kronecker delta function δ_{mn} is a function of two discrete parameters m, n. As an immediate consequence of its definition, Eq. (3.12), we have the following important property:

$$\sum_m c_m \delta_{mn} = c_n \tag{3.42}$$

where c_m is any function of the discrete variable m That is, δ_{mn} annihilates all terms in the sum for which $m \neq n$, and the only surviving term is $c_n \delta_{nn} = c_n$. Equation (3.42) may be used as the *definition* of the Kronecker delta, since its validity for all functions c_m implies the conditions of Eq. (3.12).

The Dirac delta function $\delta(x - x')$ plays the same role for functions of a continuous variable as the Kronecker delta does in the case of functions of a discrete variable. It is defined through an equation entirely analogous to (3.42):

$$\int_a^b f(x)\, \delta(x - x')\, dx = f(x'), \qquad a < x' < b \tag{3.43}$$

Here the integration sign naturally replaces the summation sign of Eq. (3.42), since x is a continuous variable. According to the definition, whatever the function $f(x)$ may be, the delta function appearing in the integral picks out the value of $f(x)$ at the single point x'; the integral does not take account of the behaviour of $f(x)$ anywhere else. Evidently, this can be the case only if

$$\delta(x - x') = 0 \quad \text{for all} \quad x \neq x' \tag{3.44a}$$

At $x = x'$, the delta function *cannot be finite*; for a function which is finite at a single point and is zero everywhere else does not 'enclose any area' under it, and its integral must vanish. Actually, from Eq. (3.43), $\int \delta(x - x')\, dx = 1$. Therefore, we conclude that

$$\delta(x, x') = \infty \quad \text{when } x = x' \tag{3.44b}$$

The form of the Dirac delta function as given by Eqs. (3.44) is very peculiar indeed. Nevertheless, it is of great utility. For ease of visualization one may think of it as the limit of a suitable sequence of more ordinary functions $\delta_\varepsilon(x - x')$ as $\varepsilon \to 0$. One of the possibilities, illustrated in Fig. 3.1 is the following:

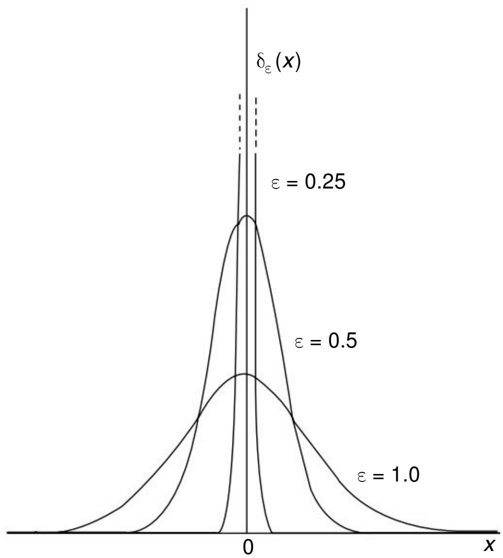

Fig. 3.1 ***The Dirac delta function as a limit of a sequence of functions $\delta_\varepsilon(x)$.***

$$\delta_\varepsilon(x - x') = (2\pi\varepsilon^2)^{-1/2} \exp\left[-(x - x')^2/2\varepsilon^2\right]$$

In the limit $\varepsilon \to 0$ it satisfies Eqs (3.44) and also the condition

$$\int \delta(x - x')\; dx = 1 \tag{3.44c}$$

as required by Eq. (3.43). Other ways of representing the delta function, and its most important properties, are listed in Appendix C.

In our studies we will have to use frequently the three-dimensional Dirac delta function $\delta(\mathbf{x} - \mathbf{x}')$. It is defined by

$$\delta(\mathbf{x} - \mathbf{x}') = \delta(x - x')\,\delta(y - y')\,\delta(z - z') \tag{3.45}$$

3.7 OBSERVABLES: COMPLETENESS AND NORMALIZATION OF EIGENFUNCTIONS

Let us recall now the observation at the end of Sec. 3.2 that if a dynamical variable is to be considered as observable, the operator representing it must be self-adjoint.

A further requirement, the origin of which is more subtle, is that the eigenfunctions of the operator should form a complete set, in a sense now to be explained. Any dynamical variable represented by a self-adjoint operator having a complete set of eigenfunctions qualifies to be called an *observable*.

Let A be a self-adjoint operator pertaining to some physical system. Its eigenfunctions $\{\phi_a\}$ are said to form a *complete set if any arbitrary wave function* ψ of the system can be 'expanded' into a linear combination,

$$\psi = \sum_{a \in D} c_a \phi_a + \int_{a \in C} c_a \phi_a \, da \tag{3.46}$$

of the members of the set. The linear combination includes summation over the discrete part of the eigenvalue spectrum (indicated by $a \in D$, D for discrete), as well as integration over the continuous part. The assumption that a set $\{\phi_a\}$ is complete, i.e. the postulate that the expansion (3.46) is possible for arbitrary ψ, is often referred to as the *expansion postulate*.

Assuming the expansion (3.46), let us evaluate the norm of ψ (taken to be 1) in terms of the coefficients c_a. Consider first the case of an operator A whose eigenvalue spectrum is wholly discrete, so that the second (integral) term is not present in Eq. (3.46). We have, on using the orthonormality property, Eq. (3.41a), of the ϕ_a,

$$\begin{aligned} 1 = \int \psi^* \psi d\tau &= \int \left(\sum_{a'} c_{a'}{}^* \phi_{a'}{}^* \right) \left(\sum_a c_a \phi_a \right) d\tau \\ &= \sum_{a'} \sum_a c_{a'}{}^* c_a \int \phi_{a'}{}^* \phi_a \, d\tau \\ &= \sum_{a'} \sum_a c_{a'}{}^* c_a \, \delta(a', a) \\ &= \sum_a c_a{}^* c_a \, \delta(a, a) \end{aligned} \tag{3.47}$$

In the last step, the sum over a' has been reduced to the single non-vanishing term, for which $a' = a$. Now, of the two possibilities corresponding to Eqs (3.41b) and (3.41c), namely $\delta(a, a) = 1$ or ∞, we must obviously reject the latter, since that would make the above equation inconsistent. Hence, we conclude that: *Eigenfunctions belonging to discrete eigenvalues are normalizable.*

Setting $\delta(a, a') = \delta_{aa'}$ in the above equation, we obtain

$$\sum_a | c_a |^2 = 1 \tag{3.48}$$

If we had a *continuous* instead of a discrete spectrum for a, integrals would appear in the place of summations in Eq. (3.47) and the penultimate expression in Eq. (3.47) would become

$$1 = \int c_a da \int c_{a'}{}^* \delta(a, a') \, da' \tag{3.49}$$

The integral over a' here vanishes if $\delta(a', a)$ is taken to be the Kronecker delta, and of course this would make Eq. (3.49) inconsistent. Hence we have to take $\delta(a, a')$

as the Dirac delta function in this case, so that: *Eigenfunctions belonging to continuous eigenvalues are of infinite norm.*

Equation (3.49) now simplifies to

$$\int | c_a |^2 \, da = 1 \tag{3.50}$$

More generally, if the spectrum of A has both discrete and continuous parts, we have

$$\sum_{a \in D} | c_a |^2 + \int_{a \in C} | c_a |^2 \, da = 1 \tag{3.51}$$

Hereafter, for economy of notation we will write out all delta functions, sums over eigenvalues, etc., as if the spectrum were discrete. For example, we abbreviate (3.46) and (3.41a) to

$$\psi = \sum_a c_a \phi_a \tag{3.52}$$

and

$$\int \phi_a{}^* \phi_{a'} \, d\tau = \delta_{aa'} \tag{3.53}$$

respectively.

Whenever the spectrum has a continuous part, the summation signs are to be understood as including integrations over the continuous part.

3.8 CLOSURE

Any set of functions $\{\phi_a\}$ which is orthonormal and complete, i.e., for which Eq. (3.53) and the expansion postulate (3.52) are valid, has the important property of *closure:*

$$\sum_a \phi_a(\mathbf{x}) \, \phi_a{}^* (\mathbf{x}') = \delta(\mathbf{x} - \mathbf{x}') \tag{3.54}$$

The proof depends on the fact that the coefficients c_a in Eq. (3.52) can be expressed as

$$c_a = \int \phi_a{}^* \, \psi \, d\tau \tag{3.55}$$

This may be verified by evaluating the right hand side using Eqs. (3.52) and (3.53). On feeding this expression back into Eq. (3.52) we get

$$\begin{aligned}\psi(\mathbf{x}) &= \sum_a c_a \phi_a(\mathbf{x}) = \sum_a \left[\int \phi_a{}^* (\mathrm{x}') \, \psi(\mathbf{x}') \, d\tau'\right] \phi_a(\mathbf{x}) \\ &= \int \left[\sum_a \phi_a(\mathbf{x}) \, \phi_a{}^*(\mathbf{x}')\right] \psi(\mathbf{x}') \, d\tau'\end{aligned}$$

The last step is simply a matter of interchanging the order of integration and summation, to carry out the summation first. The above equation states that the value of the integral is just the value, at the particular point $\mathbf{x} = \mathbf{x}'$, of the function $\psi(\mathbf{x}')$ appearing in the integrand. Evidently, the function which multiplies $\psi(\mathbf{x}')$ must be the Dirac delta function. This is just what Eq. (3.54) asserts.

The converse of what we have just proved can also be shown easily. If the set $\{\phi_a\}$ obeys the closure relation (3.54) and is orthonormal, then it ·satisfies the expansion postulate. For, on writing

$$\psi(\mathbf{x}) = \int \psi(\mathbf{x}')\, \delta(\mathbf{x} - \mathbf{x}')\, d\tau'$$

and substituting from Eq. (3.54) for the delta function, we recover the expansion (3.52) with c_a given by Eq. (3.55).

3.9 PHYSICAL INTERPRETATION OF EIGENVALUES, EIGENFUNCTIONS AND EXPANSION COEFFICIENTS

To bring out the physical meaning of eigenvalues, etc., we consider the expectation value of an observable A in some arbitrary state ψ. On expanding ψ in terms of the eigenfunctions ϕ_a of the *same* observable A, we have

$$\begin{aligned} \langle A \rangle &= \int \psi^* A\psi \, d\tau \\ &= \int (\textstyle\sum c_{a'}^* \phi_{a'}^*)\, A\, (\textstyle\sum c_a \phi_a)\, d\tau \\ &= \textstyle\sum\sum c_{a'}^* c_a \int \phi_{a'}^* A \phi_a \, d\tau \\ &= \textstyle\sum\sum c_{a'}^* c_a \, a\, \delta_{a'a} \end{aligned}$$

In the last step we have used the fact that $A\phi_a = a\phi_a$ and that the functions ϕ_a are orthonormal. Hence

$$\langle A \rangle = \sum_a |\, c_a \,|^2\, a \tag{3.56}$$

This equation states that $\langle A \rangle$ is the *weighted average* of the eigenvalues a of A. The weight factors are the positive quantities $|\, c_a \,|^2$ whose sum, according to Eq. (3.48) is unity. It is evident that the meaning of these observations is the following:

The result of any measurement of A is one of its eigenvalues. The probability that a particular value a comes out as the answer, when the system is in the state ψ, is given by[6] $|\, c_a \,|^2$ with c_a given by Eq. (3.53); when repeated measurements of A are made on systems in the state ψ the number of times the answer a is obtained is expected to be proportional to $|\, c_a \,|^2$. We see thus that *the physical significance of the eigenvalues of any observable is that they are the possible results of measurement of the observable.* The significance of the eigenfunctions can also be seen now. Suppose ψ is itself chosen to be one of the eigenfunctions of A, say $\phi_{a'}$. Eq. (3.53) tells us that in this case $c_a = \int \phi_a^* \phi_{a'}\, d\tau = \delta_{aa'}$. Thus $c_a = 1$ for $a = a'$ and zero for all other a. Hence the probability $|\, c_a \,|^2$ for getting the answer a on measuring A is unity for $a = a'$. In other words, the answer is certain to be a'. Thus *the eigenfunction $\phi_{a'}$ of A represents*

[6] If a is in the continuous part of the spectrum, then the probability that the value of A lies between a and $a + da$ is given by $|\, c_a \,|^2\, da$. Then $|\, c_a \,|^2$ is the corresponding probability *density*. Whether a is in the continuous part or is a discrete eigenvalue, c_a itself is referred to as a *probability amplitude.*

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a state in which the observable A has a definite value a′. Expressed differently, the uncertainty in the value of A is zero if the system is in one of the eigenstates of A.

Returning to the interpretation of c_a as a *probability amplitude* and $| c_a |^2$ as a probability (or probability density if a is continuous), we observe that this is exactly the same kind of interpretation which was given to the wave function $\psi(\mathbf{x})$. The only difference is that instead of the position $\mathbf{x}$ we have the values a of the observable A. Therefore, considering c_a as a function of a, we can legitimately call it the 'A-space wave function' just as $\psi(\mathbf{x})$ was called the coordinate space or configuration space wave function. The function c_a contains just the same information about the system as $\psi(\mathbf{x})$ does, since we can get $\psi(\mathbf{x})$ from c_a through Eq. (3.52); so it may be used instead of $\psi(\mathbf{x})$ for describing the state. We say that c_a and $\psi(\mathbf{x})$ are different *representations* of the state. Representation theory will be considered in detail in Chapter 7.

3.10 MOMENTUM EIGENFUNCTIONS; WAVE FUNCTIONS IN MOMENTUM SPACE

To illustrate the general properties proved in the last three sections we consider now the eigenvalue problem for the momentum operator. Consider first the one-dimensional case. The eigenvalue equation is

$$-i\hbar \frac{d\phi_{p'}(x)}{dx} = p'\phi_{p'}(x)$$

In view of the result proved in Example 3.8, p. 88, p' is the unique value of momentum which a particle has when it is in the eigenstate $\phi_{p'}(x)$. The solution of the equation is $\phi_{p'}(x) = c \exp [ip'x/\hbar]$, c being an arbitrary constant.

(*a*) *Self-adjointness and Reality of Eigenvalues:* We have seen in Example 3.7, p. 87, that $(-i\hbar\, d/dx)$ is self-adjoint only with respect to functions which either vanish as $x \to \pm\infty$ or obey periodic boundary conditions in a box. Neither of these would be satisfied by $\phi_{p'}(x)$ if p' had an imaginary part. (Verify). Thus reality of the eigenvalues p' is explicitly seen to follow as a consequence of self-adjointness.

(*b*) *Normalization and Closure:* When normalized within a (one-dimensional) box of length L, with periodic boundary conditions, the momentum eigenfunctions are given by

$$\phi_{p'}(x) = \frac{1}{\sqrt{L}} e^{ip'x/\hbar}, \quad p' = \frac{2\pi\hbar n}{L} \quad (n = 0, \pm 1, \pm 2 \ldots) \tag{3.57}$$

They are mutually orthogonal:

$$\int \phi_{p'}{}^* \phi_{p''}\, dx = \frac{1}{L}\int_{-L/2}^{L/2} e^{i(p''-p')x/\hbar}\, dx$$

$$= \frac{2\hbar \sin [(p'' - p')\, L/2\hbar]}{(p'' - p')\, L} \tag{3.58}$$

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which vanishes when $p'' \neq p'$ since $(p'' - p')\, L/2\hbar$ is an integral multiple of π. Further, the set of functions (3.57) possesses the closure property:

$$\begin{aligned}
\sum_{p'} \phi_{p'}(x)\, \phi_{p'}{}^*(x') &= \sum_{n=-\infty}^{\infty} \frac{1}{L} \exp\left[\frac{2\pi i\, n\,(x-x')}{L}\right] \\
&= \lim_{N\to\infty} \sum_{n=-N}^{N} \frac{1}{L} \exp\left[\frac{2\pi i n(x-x')}{L}\right] \\
&= \lim_{N\to\infty} \frac{1}{L} \cdot \frac{\sin[(2N+1)(x-x')\,\pi/L]}{\sin[(x-x')\,\pi/L]} \\
&= \frac{\pi}{L}\,\delta\left(\frac{\pi}{L}(x-x')\right) = \delta(x-x') \qquad (3.59)
\end{aligned}$$

(The properties of delta functions which we have used here may be found in Appendix C.) Thus the eigenfunctions of the self-adjoint operator representing momentum form a complete set. This confirms that momentum is indeed an observable in quantum mechanics. Though we have considered above only the one-dimensional case explicitly, it is obvious that the same kind of arguments hold good in three dimensions, where the box-normalized eigenfunctions of momentum $\mathbf{p}'$ are just the functions (2.30) with $\mathbf{k} = (\mathbf{p}'/\hbar)$.

If we consider momentum eigenfunctions as defined in the whole of space rather than within a box, the eigenvalues no longer form a discrete set as in Eq. (3.57). Any p' from $-\infty$ to $+\infty$ (in three dimensions, any real vector $\mathbf{p}$) is an eigenvalue of momentum. Since the eigenvalue spectrum is thus continuous, the orthonormality relation of the eigenfunctions involves the Dirac (rather than Kronecker) delta function:

$$\int \phi_{\mathbf{p}'}{}^*(\mathbf{x})\, \phi_{\mathbf{p}''}(\mathbf{x})\, d\tau = \delta\,(\mathbf{p}' - \mathbf{p}'') \qquad (3.60)$$

The momentum eigenfunctions with this normalization are

$$\phi_{\mathbf{p}'}(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}}\, e^{i\mathbf{p}'\cdot\mathbf{x}/\hbar} \qquad (3.61)$$

To verify this statement we substitute Eq. (3.61) in Eq. (3.60) and take the integrals with respect to x, y and z each from $-\frac{1}{2}L$ to $+\frac{1}{2}L$, and finally make $L \to \infty$. The triple integral evidently reduces to a product of three single integrals. A typical one is

$$\begin{aligned}
&\lim_{L\to\infty} \int_{-L/2}^{L/2} \left(\frac{1}{\sqrt{2\pi\hbar}} \cdot e^{ip_x{}'x/\hbar} \cdot \frac{1}{\sqrt{2\pi\hbar}}\, e^{ip_x{}''x/\hbar}\right) dx \\
&= \lim_{L\to\infty} \frac{1}{(2\pi\hbar)} \frac{2\hbar}{(p_x{}' - p_x{}'')} \cdot \sin\left(\frac{(p_x{}' - p_x{}'')\,L}{2\hbar}\right) \\
&= \frac{1}{2\hbar}\,\delta\left(\frac{(p_x{}' - p_x{}'')}{2\hbar}\right) = \delta\,(p_x{}' - p_x{}'')
\end{aligned}$$

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Similarly the y and z integrals yield $\delta(p_y' - p_y'')$ and $\delta(p_z' - p_z'')$. On taking the product of all these we obtain $\delta(\mathbf{p}' - \mathbf{p}'')$, thus confirming that the eigenfunctions of Eq. (3.61) are normalized according to Eq. (3.60). The closure property for these functions takes the form

$$\int \phi_{\mathbf{p}}(\mathbf{x})\, \phi_{\mathbf{p}}{}'^{*}(\mathbf{x}')\, d^3p' = \delta(\mathbf{x} - \mathbf{x}') \tag{3.62}$$

It is usually the practice to write the momentum eigenfunctions in terms of the propagation vector $\mathbf{k}'$ as

$$\phi_{\mathbf{k}'}(\mathbf{x}) = \frac{1}{(2\pi)^{3/2}}\, e^{i\mathbf{k}'.\mathbf{x}} \tag{3.63}$$

to save writing factors $\hbar$. Here $\mathbf{k}' = \mathbf{p}'/\hbar$. Note that the normalization constant here differs from that in Eq. (3.61) by a factor $\hbar^{3/2}$. It may be easily verified that with this normalization

$$\int \phi_{\mathbf{k}'}{}^{*}(\mathbf{x})\, \phi_{\mathbf{k}''}(\mathbf{x})\, d\tau = \delta(\mathbf{k}' - \mathbf{k}'') \tag{3.64a}$$

$$\int \phi_{\mathbf{k}}(\mathbf{x})\, \phi_{\mathbf{k}}{}^{*}(\mathbf{x}')\, d^3k = \delta(\mathbf{x} - \mathbf{x}') \tag{3.64b}$$

(c) *The Wave Function and Operators in Momentum Space:* Since the momentum eigenfunctions form a complete orthonormal set, we can expand any arbitrary wave function $\psi(\mathbf{x})$ in terms of them as

$$\psi(\mathbf{x}) = \int c(\mathbf{p})\, \phi_{\mathbf{p}}(\mathbf{x})\, d^3p = \frac{1}{(2\pi\hbar)^{3/2}} \int c(\mathbf{p}) e^{i\mathbf{p}.\mathbf{x}/\hbar}\, d^3p \tag{3.65a}$$

which is a special case of Eq. (3.52). Conversely, from Eq. (3.55) we get,

$$c(\mathbf{p}) = \int \phi_{\mathbf{p}}{}^{*}(\mathbf{x})\, \psi(\mathbf{x})\, d^3x = \frac{1}{(2\pi\hbar)^{3/2}} \int \psi(\mathbf{x})\, e^{-i\mathbf{p}.\mathbf{x}/\hbar} d^3x \tag{3.65b}$$

Equivalently, we have in terms of the propagation vector $\mathbf{k} = \mathbf{p}/\hbar$,

$$\psi(\mathbf{x}) = \frac{1}{(2\pi)^{3/2}} \int c(\mathbf{k}) e^{i\mathbf{k}.\mathbf{x}}\, d^3k, \quad c(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int \psi(\mathbf{x}) e^{-i\mathbf{k}.\mathbf{x}} d^3x \tag{3.66}$$

where

$$c(\mathbf{k}) = \hbar^{3/2}\, c(\mathbf{p}) \tag{3.67}$$

It is evident that $\psi(\mathbf{x})$ is the *Fourier transform* of $c(\mathbf{k})$.

The physical interpretation of $c(\mathbf{p})$ follows from the discussion of Sec. 3.9: $c(\mathbf{p})$ gives the probability amplitude, and $|c(\mathbf{p})|^2$ the probability density, for the momentum to be in the neighbourhood of $\mathbf{p}$. The expectation value of any function of momentum can, therefore, be expressed as

$$\langle f(\mathbf{p}) \rangle = \int \left| c(\mathbf{p}) \right|^2 f(\mathbf{p})\, d^3p \tag{3.68}$$

As indicated at the end of Sec. 3.9, $c(\mathbf{p})$ may be called the *momentum space wave function*; it contains the same information as the configuration space wave function

$\psi(\mathbf{x})$, in view of Eq. (3.65a), and is an equally valid representation of the state of the system. In momentum space, the dynamical variables would be represented by operators which act on $c(\mathbf{p})$. In particular, position and momentum are represented by

$$\mathbf{x}_{op} = i\hbar\nabla_{\mathbf{p}}, \quad \mathbf{p}_{op} = \mathbf{p} \tag{3.69}$$

where $\nabla_{\mathbf{p}} = (\partial/\partial p_x, \partial/\partial p_y, \partial/\partial p_z)$. In other words the momentum space wave functions corresponding to $\mathbf{x}\ \psi(\mathbf{x})$ and $-i\hbar\nabla\psi(\mathbf{x})$ are $i\hbar\nabla_{\mathbf{p}}c(\mathbf{p})$ and $\mathbf{p}\ c(\mathbf{p})$ respectively. This is easily verified by noting that on acting with $-i\hbar\nabla$ on Eq. (3.65a), $c(\mathbf{p})$ in the integral gets replaced by $\mathbf{p}\ c(\mathbf{p})$ and further that to replace $\psi(\mathbf{x})$ by $\mathbf{x}\psi(\mathbf{x})$ what one has to do is to operate on $c(\mathbf{p})$ with $i\hbar\nabla\mathbf{p}$ as is evident from Eq. (3.65b).

It is important to note that though the explicit forms of the operators in momentum space are necessarily different from those in configuration space, the commutation relations given by Eqs. (3.13) hold in both cases.

Example 3.11 What is the probability distribution of momentum of a particle with the Gaussian wave function $\psi(x) = \left(\sigma.\sqrt{\pi}\right)^{-1/2}.\exp[-x^2/2\sigma^2]$? Specializing the above treatment to the one-dimensional case, we have

$$c(k) = \frac{1}{(2\pi)^{1/2}}\int \psi(x)\, e^{-ikx} dx.$$

This integral can be evaluated by expanding e^{-ikx} in series and using the fact that

$$\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2}.\, x^{2s}\, dx = (2\pi)^{1/2}\frac{(2s)!}{s!}\frac{\sigma^{2s+1}}{2^s}, \quad (s = 0, 1, 2, ...)$$

(This may be inferred from the integral evaluated in Example 3.6, p. 85). Since integrals involving odd powers (x^{2s+1}) vanish, we have

$$\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} e^{-ikx} dx = (2\pi)^{1/2}\sum_{s=0}^{\infty}\frac{(2s)!}{s!}\frac{\sigma^{2s+1}}{2^s}\frac{(-ik)^{2s}}{(2s)!} = (2\pi)^{1/2}\,\sigma e^{-\frac{1}{2}\sigma^2 k^2},$$

and hence

$$c(k) = \left(\sigma/\sqrt{\pi}\right)^{1/2}\exp\left[-\frac{1}{2}\sigma^2 k^2\right].$$

The probability density function for momentum is $\left|c(p)\right|^2 = \hbar^{-1}\left|c(k)\right|^2$. It is noteworthy that the Gaussian wave function has a Gaussian form in momentum space too. (Compare the widths of the wave packets $\psi(x)$ and $c(p)$). The relation $\psi(x) = (2\pi)^{-1/2}\int c(k)e^{ikx}\, dk$ now becomes

$$\left(\frac{1}{\sigma\sqrt{\pi}}\right)^{1/2} e^{-x^2/2\sigma^2} = \frac{1}{(2\pi)^{1/2}}\left(\frac{\sigma}{\sqrt{\pi}}\right)^{1/2}\int e^{-\frac{1}{2}\sigma^2 k^2} e^{ikx}\, dk$$

■

Example 3.12 As an example in three-dimensions, consider the ground state wave function of the hydrogen atom (See Sec. 4.16):

$$\psi(\mathbf{x}) = \left(\frac{1}{\pi a_0{}^3}\right)^{1/2} e^{-r/a_0}$$

It depends only on the radial distance r. The constant a_0 is the radius of the first Bohr orbit. In momentum space, we have

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$$c(\mathbf{p}) = \left(\frac{1}{2\pi\hbar}\right)^{3/2}\left(\frac{1}{\pi a_0^{\,3}}\right)^{1/2}\int e^{-r/a_0}.\,e^{-i\mathbf{p}.\mathbf{x}/\hbar}\,d\tau.$$

It is advantageous to transform to spherical polar coordinates, choosing the polar axis along the direction of **p**. Then $\mathbf{p.x} = pr\cos\theta$, and the integrand is independent of the azimuthal angle φ. We thus have

$$c(\mathbf{p}) = (8\pi^4\hbar^3 a_0^{\,3})^{-1/2}\int_0^{\infty}\int_0^{\pi}\int_0^{2\pi}\exp\left[-\frac{r}{a_0}-\frac{ipr\cos\theta}{\hbar}\right]d\varphi\sin\theta\,d\theta.r^2dr$$

On carrying out the integrations we get

$$c(\mathbf{p}) = \frac{1}{\pi}.\left(\frac{2a_0}{\hbar}\right)^{3/2}.\frac{1}{\left[\left(p^2a_0^{\,2}/\hbar^2\right)+1\right]^2}.$$

The probability that the magnitude and direction lie within $(p, p+dp)$, $(\alpha, \alpha+d\alpha)$, $(\beta, \beta+d\beta)$ is *not* $|c(\mathbf{p})|^2\,dp\,d\alpha\,d\beta$ but $|c(\mathbf{p})|^2\,p^2dp\sin\alpha\,d\alpha\,d\beta$. If the probability distribution of only one of the variables is of interest, integrate over the others. For instance, in the present case where $c(\mathbf{p})$ is independent of α, β we get $P(p)\,dp = |c(\mathbf{p})|^2\,4\pi p^2\,dp$. ■

3.11 THE UNCERTAINTY PRINCIPLE

Whenever we have some quantity which can assume several values with various probabilities, it is customary to use the mean squared deviation as a measure of the 'width' of the probability distribution or in other words, the uncertainty in the value of the quantity. The uncertainty in the values of quantum mechanical observables also is defined in the same way. If A is an observable and $\langle A\rangle$ its expectation or mean value in the state ψ,

$$\mathcal{A} \equiv A - \langle A\rangle \tag{3.70}$$

is the self-adjoint operator representing the deviation of A from its mean. The expectation value of $\mathcal{A}^2$ is the mean squared deviation; and the *uncertainty* in A (usually denoted by ΔA) is defined by

$$(\Delta A)^2 = \left\langle \mathcal{A}^2\right\rangle \tag{3.71a}$$

$$\equiv \left\langle\left(A-\langle A\rangle\right)^2\right\rangle = \left\langle A^2\right\rangle - \langle A\rangle^2 \tag{3.71b}$$

The uncertainty depends on the state ψ in which the system is.

We can always find states in which ΔA vanishes; in fact we know that *the eigenfunctions ϕ_a of A represent such states* (See Sec. 3.9).

Now we ask: Given *two* observables A, B, can we find some state in which *both* ΔA and ΔB are zero, so that both A and B have precise values? We already know that at least in one case (position and momentum) the answer is in the negative. To study this question in more general terms we make essential use of the positivity property (3.33). We apply it to the operator $\mathcal{A} - i\lambda\mathcal{B}$ where $\mathcal{A}$ is defined by Eq. (3.70); $\mathcal{B}$ is defined similarly as $\mathcal{B} = B - \langle B\rangle$ and λ is a real parameter (not an operator).

Noting that the adjoint of $(\mathcal{A} - i\lambda\mathcal{B})$ is $(\mathcal{A} + i\lambda\mathcal{B})$ since $\mathcal{A}$ and $\mathcal{B}$ are self-adjoint, we have from Eq. (3.33)

$$\langle(\mathcal{A} - i\lambda\mathcal{B})\,(\mathcal{A} + i\lambda\mathcal{B})\rangle \geq 0 \tag{3.72}$$

Expanding the product, keeping in mind that $\mathcal{A}$ and $\mathcal{B}$ may not commute, we simplify the expression in Eq. (3.72):

$$\langle\mathcal{A}^2\rangle + \lambda^2\langle\mathcal{B}^2\rangle - \lambda\langle\mathcal{C}\rangle \geq 0 \tag{3.73}$$

Here $\mathcal{C}$ is the self-adjoint operator defined by

$$i\mathcal{C} = [\mathcal{A}, \mathcal{B}] = [A, B] \tag{3.74}$$

The left hand side of (3.73) has its minimum value when its derivative with respect to λ is zero. This happens when

$$\lambda = \frac{\langle\mathcal{C}\rangle}{2\langle\mathcal{B}^2\rangle} \tag{3.75}$$

For this value of λ Eq. (3.73) becomes

$$\langle\mathcal{A}^2\rangle - \frac{\langle\mathcal{C}^2\rangle}{4\langle\mathcal{B}^2\rangle} \geq 0 \tag{3.76}$$

Hence we have $\langle\mathcal{A}^2\rangle\langle\mathcal{B}^2\rangle \geq \frac{1}{4}\langle\mathcal{C}^2\rangle$ or, in view of the definition (3.71a) of the uncertainty, and Eq. (3.74) for $\mathcal{C}$;

$$(\Delta\mathcal{A})^2\,(\Delta\mathcal{B})^2 \geq -\tfrac{1}{4}\langle[A, B]\rangle^2 \tag{3.77}$$

Thus a lower limit on the product of the uncertainties in A and B is set by the minimum value possible for the expectation value of the commutator $[A, B]$. *Equation (3.77) gives the general statement of the uncertainty principle for any pair of observables A, B.*

In particular, if A, B are a *canonically conjugate* pair of operators, which are characterized by

$$[A, B] = i\hbar \tag{3.78}$$

then

$$(\Delta A)^2\,(\Delta B)^2 \geqslant \tfrac{1}{4}\hbar^2, \quad \text{or} \quad (\Delta A)\;(\Delta B) \geqslant \tfrac{1}{2}\hbar \tag{3.79}$$

Any position coordinate variable x and its canonically conjugate momentum variable p of course satisfy Eq. (3.78) and, therefore, we have

$$(\Delta x)\,(\Delta p) \geqslant \tfrac{1}{2}\hbar \tag{3.80}$$

This is the precise statement of the position-momentum uncertainty principle. In three dimensions,

$$(\Delta x)\;(\Delta p_x) \geqslant \tfrac{1}{2}\hbar,\; (\Delta y)\;(\Delta p_y) \geqslant \tfrac{1}{2}\hbar,\; (\Delta z)\,(\Delta p_z) \geqslant \tfrac{1}{2}\hbar \tag{3.81}$$

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It may be recalled that a qualitative version of it was deduced from heuristic arguments in Sec. 1.14.

We have seen in Example 3.5 (p. 83) that the azimuthal angle φ is not a proper dynamical variable[7] in quantum mechanics. Hence, even though a commutation rule $[\varphi, L_z] = i\hbar$ of the form (3.78) can be formally written down, we cannot talk of an uncertainty relation $(\Delta\varphi)\ (\Delta L_z) \geqslant \frac{1}{2}\hbar$ since $\Delta\varphi$ itself cannot be properly defined.

As far as the energy and time variables are concerned, one cannot talk of a Δt at a single instant of time, nor is the operator $i\hbar\left(\partial/\partial t\right)$, associated with E by Eq. (2.14), definable on an instantaneous wave function. So, even though we do have $[i\hbar\, \partial/\partial t, t] = i\hbar$, it cannot be employed in the above analysis which is based on expectation values with respect to wave functions at a single instant. However, a relation $(\Delta E)\ (\Delta t) \geqslant \hbar$ does exist. Its interpretation, which is quite different from that of Eq. (3.80) will be given elsewhere.

Example 3.13 As an example involving a pair of observables which are *not* canonically conjugate, let us consider the following question: Does there exist a state ϕ in which $\Delta L_x = \Delta L_y = 0$, where $\mathbf{L}$ is the angular momentum vector? This question is equivalent to asking whether there is any state ϕ which is an eigenstate of both L_x and L_y: $L_x\phi = m_x\phi$, $L_y\phi = m_y\phi$ where m_x, m_y are constants. If such a state exists, $L_xL_y\phi = L_xm_y\phi = m_xm_y\phi = L_yL_x\phi$. Using this result and the commutation relation $[L_x, L_y] = i\hbar L_z$ proved in Example 3.4 (p. 81), one finds $L_z\phi = (i\hbar)^{-1}\left(L_xL_y\phi - L_yL_x\phi\right) = 0$. So ϕ must belong to the zero eigenvalue of L_z. Repeating the above arguments, now starting with $L_y\phi = m_y\phi$ and $L_z\phi = 0$, one finds $m_x = 0$ and similarly $m_y = 0$. Thus if more than one component of angular momentum is to have a definite value in a state ϕ, it must satisfy $L_x\phi = L_y\phi = L_z\phi = 0$ and hence $\mathbf{L}^2\phi = 0$, where $\mathbf{L}^2 = L_x^2 + L_y^2 + L_z^2$. Such a state does exist: See Υ_{00} of Eq.(4.90) together with polar coordinate expressions (4.51) for L_x, L_y, L_z. ■

3.12 STATES WITH MINIMUM VALUE FOR UNCERTAINTY PRODUCT

Whether the equality or inequality holds in (3.77) depends on the state ψ in which the system is. The minimum value of $(\Delta A)\ (\Delta B)$ occurs evidently in those states for which the equality sign holds in (3.77) and hence in (3.72) with λ given by Eq. (3.75). But Eq. (3.33a) applied to the present case shows that the left hand side of (3.72) can vanish only in states ψ for which $(\mathcal{A} + i\lambda\mathcal{B})\,\psi$ vanishes, i.e.

$$\left(\mathcal{A} + \frac{i\langle C\rangle}{2\langle\mathcal{B}^2\rangle}\cdot\mathcal{B}\right)\psi = 0 \tag{3.82}$$

The case of greatest interest is when A and B are canonically conjugate position and momentum variables x and p. Then

$$\mathcal{A} = x - \langle x\rangle,\ \mathcal{B} = p - \langle p\rangle,\ \langle\mathcal{B}^2\rangle = (\Delta p)^2,$$
$$C = -i\,[A, B] = \hbar \tag{3.83}$$

Using the explicit differential operator form for B, we rewrite Eq. (3.82) in this case as

[7] For a review on phase variables and operators to be associated with them in quantum mechanics, *see* P. Carruthers and M. M. Nieto, *Rev. Mod. Phys.*, **40**, 411, 1968.

$$\left(x - \langle x \rangle\right)\psi + \frac{i\hbar}{2(\Delta p)^2}\left(-i\hbar\frac{d}{dx} - \langle p \rangle\right)\psi = 0$$

or

$$\frac{d\psi}{dx} + \left\{\frac{2(\Delta p)^2}{\hbar^2}\left(x - \langle x \rangle\right) - \frac{i\langle p \rangle}{\hbar}\right\}\psi = 0 \tag{3.84}$$

where any desired value may be assigned to Δp. It is a trivial matter to verify that the solution of this equation is

$$\psi = N \exp\left[-\frac{(\Delta p)^2}{\hbar^2}.\left(x - \langle x \rangle\right)^2 + \frac{i\langle p \rangle x}{\hbar}\right] \tag{3.85a}$$

The wave function is normalized, i.e. $\int|\psi|^2\,dx = 1$, if N is chosen as

$$N = \left[\frac{2(\Delta p)^2}{\pi\hbar^2}\right]^{1/4} \tag{3.85b}$$

Equations (3.85) give the normalized wave functions for which $(\Delta x)\,(\Delta p)$ has the minimum value $\frac{1}{2}\,\hbar$. Note that ψ has the form of a Gaussian function 'modulated' by the oscillatory factor $\exp\,[i\langle p\rangle\, x/\hbar]$. The latter, being of unit modulus, does not affect the probability density $|\psi|^2$ which is Gaussian with maximum at $x = \langle x\rangle$:

$$|\psi|^2 = \left[2\pi(\Delta x)^2\right]^{-\frac{1}{2}}.\exp\left[-\frac{\left(x - \langle x \rangle\right)^2}{2(\Delta x)^2}\right] \tag{3.86}$$

(The normalization may be checked by comparison of this expression with Example 3.6 p. 85.) Since $|\psi|^2$ is negligibly small outside a region having dimensions of the order $\Delta x = \left(\frac{1}{2}\,\hbar/\Delta p\right)$, the wave function (3.85a) is said to describe a minimum uncertainty *wave 'packet'*.

3.13 COMMUTING OBSERVABLES; REMOVAL OF DEGENERACY

In the case of two commuting observables A, B, the right hand side of the general uncertainty relation (3.77) vanishes, so that states in which ΔA and ΔB are *both* zero can exist. In other words, there exist states in which both A and B have precise values and which are, therefore, eigenstates of A and B simultaneously.

Consider an eigenstate ϕ_a of A. When $AB = BA$, we have

$$AB\phi_a = BA\phi_a = B.a\phi_a,$$

i.e.

$$A\,(B\phi_a) = a\,(B\phi_a) \tag{3.87}$$

Thus not only ϕ_a but $B\phi_a$ also is an eigenstate of A belonging to the *same* eigenvalue a. If a happens to be a non-degenerate eigenvalue, there is (by definition) only one eigenfunction belonging to it and hence $B\phi_a$ must be a constant multiple of ϕ_a, say

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$$B\phi_a = b\phi_a \tag{3.88}$$

But this means that ϕ_a is also an eigenfunction of B, belonging to the eigenvalue b. This information may be included in the notation for the eigenfunction by renaming it as ϕ_{ab}.

Thus any eigenfunction belonging to a *non-degenerate* eigenvalue of either of a pair of commuting operators A, B is necessarily an eigenfunction of the other operator too.

What if a happens to be a degenerate eigenvalue? Then if ϕ_a is an *arbitrary* one of the (infinite number of) eigenfunctions belonging to a, all we can say from Eq. (3.87) is that $B\phi_a$ is also *one of the* eigenfunctions belonging *to a.* But it is always possible to choose a basic set of r eigenfunctions (where r is the degree of degeneracy of the eigenvalue a) in such a way that each of them is an eigenfunction of B too.[8] In this manner one can obtain a *complete set of simultaneous eigenfiunctions* ϕ_{ab} for any pair of commuting observables A, B.

It may be observed that the label b serves to distinguish between different eigenfunctions belonging to any degenerate eigenvalue a of A. If the r independent eigenfunctions belonging to a given degenerate eigenvalue a are characterized by *r distinct* values of b, then we say that the degeneracy of a is *completely removed.* This is because for that value of a, by specifying a particular value of b a particular eigenfunction is uniquely identified.

Suppose now that the degeneracy of every eigenvalue a is not removed completely by the introduction of the additional label b. Then one can introduce one or more further observables C, ... which commute with one another and with A and B, till the set of eigenvalues a, b, c, ... identify the state ϕ_{abc} ... unambiguously. That is to say, ϕ_{abc} ... and $\phi_{a'b'c'}$... are identical if and only if $a = a'$, $b = b'$, $c = c'$, ... When this is accomplished, we say that *A, B, C,* ... form a *complete commuting set of observables.* It is obvious that if each ϕ_{abc} ... is individually normalized, the set of all such simultaneous eigenfunctions forms an orthonormal set:

$$\int \phi^*_{abc}\ldots\phi_{a'b'c'}\ldots d\tau = \delta_{aa'}\,\delta_{bb'}\,\delta_{cc'}\ldots \tag{3.89}$$

For, if any two similar indices, e.g. b, b' are unequal, it means that the two wave functions belong to *distinct* eigenvalues of the corresponding operator B, and must, therefore, be orthogonal as shown in Sec. 3.5.

Since no degeneracy exists when all the eigenvalues a, b, c, ... are specified, the considerations of Sec. 3.4–3.8 can be made completely general by using the ϕ_{abc} ... in the place of the ϕ_a. Apart from the appearance of a number of indices and consequent changes in wording of some of the statements, no further complication arises. The completeness postulate, for instance, becomes $\psi = \sum_{abc\ldots} c_{abc\ldots}\phi_{abc\ldots}$, and $|c_{abc\ldots}|^2$ is now the *joint* probability[9] that the observables A, B, C, ... have the respective values a, b, c, ... etc.

[8] The proof of this statement will not be given at this stage since it involves the use of matrix concepts to be introduced later (Sec. 7.6).

[9] Or joint probability *density*, if any of the indices belongs to a continuum.

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3.14 EVOLUTION OF SYSTEM WITH TIME; CONSTANTS OF THE MOTION

The consideration of general properties of quantum systems up to this point have been concerned with various aspects of the description at any given instant of time. The basic picture of the changes which take place as time progresses is given, in the Schrödinger form of quantum mechanics, by the following postulate.

POSTULATE 4 – The state ψ varies with time in a manner determined by the Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = H_{op}\psi \tag{3.90}$$

where H_{op} is the Hamiltonian operator.[10] The basic dynamical variables $\mathbf{x}$ and $\mathbf{p}$ (or more precisely, the operators representing them) do not change with time.

However, there is nothing to prevent us from considering operators $A_{op}(\mathbf{x}, \mathbf{p}; t)$ which are explicitly time dependent. In such cases

$$\frac{d}{dt} A_{op}(\mathbf{x}, \mathbf{p}; t) = \frac{\partial}{\partial t} A_{op}(\mathbf{x}, \mathbf{p}, t) \tag{3.91}$$

i.e. the change in A_{op} is solely due to the explicit time variable since $\mathbf{x}_{op}$ and $\mathbf{p}_{op}$ are time independent.

Let us now see how expectation values change, according to the above postulate. We have

$$\frac{d}{dt}\langle A(\mathbf{x}, \mathbf{p}; t)\rangle = \frac{d}{dt}\int \psi^* A_{op}\psi d\tau$$

$$= \int\left[\frac{\partial \psi^*}{\partial t} A_{op}\psi + \psi^*\left(\frac{\partial A_{op}}{\partial t}\right)\psi + \psi^* A_{op}\frac{\partial \psi}{\partial t}\right]d\tau$$

$$= \int\left[-\frac{1}{i\hbar}\left(H_{op}\ \psi\right)^* A_{op}\psi + \psi^*\frac{\partial A_{op}}{\partial t}\psi + \frac{i}{i\hbar}\psi^* A_{op}\ H_{op}\psi\right]d\tau \tag{3.92}$$

In the last step we have used the Schrödinger equation and its conjugate, $-i\hbar\,\partial\psi^*/\partial t = (H_{op}\psi)^*$. Now, since the energy is an observable, H_{op} is a self-adjoint operator. This fact enables us to reduce the above equation to

$$\frac{d\langle A\rangle}{dt} = \int \psi^*\left\{\frac{1}{i\hbar}\left[A_{op}, H_{op}\right] + \frac{\partial A_{op}}{\partial t}\right\}\psi d\tau \tag{3.93}$$

Thus the rate of change of the *expectation value* of any dynamical variable A may be obtained as the expectation value of $(i\hbar)^{-1}\left[A, H\right] + \partial A/\partial t.$ This operator may be taken to represent the dynamical variable (dA/dt):

[10] The subscript 'op' is temporarily revived here, because the distinction between the dynamical variable as a physical concept on the one hand and the operator representing it on the other hand, is important for the following discussion.

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$$\left(\frac{dA}{dt}\right)_{op} = \frac{1}{i\hbar}\left[A_{op}, H_{op}\right] + \frac{\partial A_{op}}{\partial t} \tag{3.94}$$

It is important to note the distinction between this operator for (dA/dt), and the rate of change of A_{op} which is given by Eq. (3.91).

It may be observed in particular that Eq. (3.92) contains the statement that the norm of ψ is conserved. For, the norm is nothing but the expectation value of unity, and its time derivative is trivially seen to be zero by substituting $A = 1$ in Eq. (3.93). We have used only the hermiticity (self-adjointness) of the Hamiltonian in the proof of this result. The hermiticity of H is thus linked to conservation of probability.

Finally we observe that if A is any dynamical variable which is not explicitly time-dependent ($dA_{op}/dt = 0$) *and it commutes with* H, then $\langle A\rangle$ is independent of time, whatever ψ may be. Any such A is said to be a *conserved quantity* or *constant of the motion.* Whenever it is desired to choose a complete commuting set of observables (See Sec. 3.13) for any system, it is standard practice to take H as one of them, so that all the observables of the set will be constants of the motion because of their commutability with H. Thus if at some instant of time a system is in a state wherein these observables have specific values, i.e. in a simultaneous eigenstate of the observables, it remains in the same state for all time. The eigenvalues or related quantum numbers which label the state are *good quantum numbers* for the system, since the values of the labels of such a state remain good (unaltered) for all time.

Example 3.14 It has been noted in Example 3.5, (p. 83) that the operator for L_z in polar coordinates is $L_z = -i\hbar\ \partial/\partial\varphi$. Therefore, L_z commutes with any operator which does not depend on φ. In particular it commutes with ∇^2 which is independent of φ as may be seen from its polar coordinate form (See Example 3.1, p. 76). Therefore L_z is a constant of the motion for a free particle, whose Hamiltonian is $-\ (\hbar^2/2m)\ \nabla^2$. In fact, it is a constant of the motion even if a potential V is present, provided V is independent of φ, i.e. symmetric with respect to the z axis. Similarly L_x, L_y are constants of the motion if V is symmetric about the x and y axes respectively. If the potential is spherically symmetric, depending on r only and not on θ or φ then all components of the angular momentum are conserved. ■

Example 3.15 Show that the mean value $\langle T\rangle$ of the kinetic energy part of the Hamiltonian is equal to (1/2) $\langle(\mathbf{x}\cdot\nabla V)\rangle$ in any stationary state. What is the mean value if $V = cr^k$?

To provide a proof, begin with the observation that a stationary state wave function has the time dependence $\psi(\mathbf{x}, t) = u_n(\mathbf{x})\, e^{-iE_n t/\hbar}$, and $\psi^*(\mathbf{x}, t) = u_n^*(\mathbf{x})\, e^{iE_n t/\hbar}$. This means that in calculating the expectation value of any operator A in such a state, the two exponential factors cancel each other out, and hence $\langle A\rangle = \int u_n^*(\mathbf{x})\, Au_n(\mathbf{x})$ is a constant if the operator A is time independent, i.e., if $\partial A/\partial t = 0$. Consequently $\langle\partial A/\partial t\rangle = 0$, and by virtue of Eq. (3.86), $\langle[A, H]\rangle = 0$.

Consider now the commutator $[A, H]$ for the operator $A = \mathbf{x}\cdot\mathbf{p} = \sum_j x_j p_j$:

$$\sum_j \left[x_j p_j, p^2/2m + V(\mathbf{x})\right] = \sum_j \left\{\left[x_j p_j, \left(p^2/2m\right)\right] p_j + x_j\left[p_j, V(\mathbf{x})\right]\right\} \tag{E3.1}$$

One sees from the commutation relation $[x_j, p_k] = i\hbar\delta_{jk}$ that the first term on the right hand side of the above equation becomes $i\hbar(p_j/m)p_j = i\hbar\mathbf{p}^2/m = 2T$. The second term of the expression above becomes $(-i\hbar)\sum_j x_j\,\partial_j V = (-i\hbar)(\mathbf{x}\cdot\nabla V)$, since $[\mathbf{p}, f(\mathbf{x})] = -i\hbar\nabla f(\mathbf{x})$ for any function f, as we have already seen.

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$$[\mathbf{x}\cdot\mathbf{p}, H] = i\hbar(2T - \mathbf{x}\cdot\nabla V) \tag{E3.2}$$

Vanishing of the expectation value $\langle [A, H]\rangle$ leads now to the result that

$$\langle 2T\rangle - \langle \mathbf{x}\cdot\nabla V\rangle = 0 \tag{E3.3}$$

If $V(\mathbf{x})$ is a spherically symmetric potential, $\mathbf{x}\cdot\nabla V = \mathbf{x}\cdot(dV/dr)(\mathbf{x}/r) = r dV/dr$. In particular, if $V(r) = cr^k$, $r dV/dr = ckr^k = kV$. The last equation requires then that

$$\langle T\rangle = \frac{k}{2}\langle V\rangle \tag{E3.4}$$

In the important case of the potential of the nucleus of the hydrogen atom, $V = -e^2/r$, $k = -1$ and the above result implies that $\langle T\rangle = -(1/2)\langle V\rangle$. We also know that $\langle (T+V)\rangle = \langle H\rangle = E_n$ where E_n is the eigenvalue to which the state belongs. On combining the last two results, we obtain

$$\langle T\rangle = -\frac{1}{2}\langle V\rangle = -E_n \tag{E3.5}$$

■

3.15 NON-INTERACTING AND INTERACTING SYSTEMS

Suppose we have two systems, each with its own set of basic dynamical variables (coordinates and conjugate momenta). The two 'systems' may, for example, be two individual particles, or a particle and an atom, or two atoms, etc. Let us use the symbol '1' to denote collectively the dynamical variables of the first system, and '2' for the other. Any of the variables 1 will commute with the variable 2, because they pertain to different systems. The Hamiltonian of the *combined* system will depend on both the sets of variables, and we denote it by $H(1, 2)$. If it can be expressed as a sum of two parts

$$H(1, 2) = H_1(1) + H_2(2) \tag{3.95}$$

where either part depends on the variables of one system only, as indicated, then the two systems are *non-interacting;* they do not influence each other. This is evident from the fact that if system 1 is in an eigenstate $u(1)$ of $H_1(1)$ and system 2 is in an eigenstate $v(2)$ of $H_2(2)$ at some instant of time, this state of affairs persists for all time. For $H_1(1)$ and $H_2(2)$ commute with each other and hence with $H(1, 2)$, and so their eigenvalues are constants of the motion. Note that if

$$H_1(1)\, u(1) = E' u(1) \quad \text{and} \quad H_2(2)\, v(2) = E'' v(2), \tag{3.96}$$

then

$$\begin{aligned} H(1, 2)\, u(1)\, v(2) &= [H_1(1)\, u(1)]\, v(2) + u(1)\, [H_2(2)\, v(2)] \\ &= E\, u(1)\, v(2) \end{aligned} \tag{3.97}$$

with $E = E' + E''$.

Thus the eigenfunctions of the combined system consisting of two non-interacting subsystems are *products* of the eigenfunctions of the individual subsystems, while the energy eigenvalue E is a *sum* of E' and E''. It may be observed that operators of one system have no effect on operators or wave functions of the other. Therefore in products, we can freely change the order of occurrence of the quantities of one system relative to the other.

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If the systems 1 and 2 interact mutually, then $H(1, 2)$ cannot be separated as before; the presence of the interaction is usually manifested as an additional term $H'(1, 2)$ depending on the variables of both:

$$H(1, 2) = H_1(1) + H_2(2) + H'(1, 2) \tag{3.98}$$

Then one refers to $H'(1, 2)$ as the *interaction part* of the Hamiltonian, and $H_1(1) + H_2(2)$ as the '*free*' part. When H' is present, one no longer has eigenstates of $H(1, 2)$ as simple products $u(1)\, v(2)$. Viewed another way, if the system at some instant of time has the wave function $u(1)\, v(2)$ which is an eigenfunction of the 'free' part of $H(1, 2)$, at a later time the wave function will not be an eigenfunction of the 'free' part but a superposition of several such eigenfunctions. The determination of the manner in which the superposition develops as a consequence of interaction is one of the major problems in the application of quantum mechanics.

3.16 SYSTEMS OF IDENTICAL PARTICLES

Finally, we establish certain important properties of systems consisting of identical particles, and incidentally, provide an illustration of operators which are not definable as functions of **x** and **p**.

(*a*) *Interchange of particles; Symmetric and antisymmetric wave functions:* By definition, if two particles are *identical*, no observable effects whatever can arise from interchanging them. More precisely, *all observable quantities* must remain unaltered if the position, momentum and other dynamical variables such as spin of the first particle (which we collectively denote by 1) are interchanged with those of the second particle (collectively denoted by 2). In particular, this should be true of the Hamiltonian:

$$H(1, 2) = H(2, 1) \tag{3.99}$$

Now let us consider the *exchange operator* $\mathcal{P}$ which is *defined* by saying that it has the following effect on an arbitrary wave function $\psi(1, 2)$:

.
$$\mathcal{P}\psi(1, 2) \equiv \psi(2, 1) \tag{3.100}$$

By operating with $\mathcal{P}$ on this equation, we obtain

$$\mathcal{P}^2\psi(1, 2) = \mathcal{P}.\,\mathcal{P}\psi(1, 2) = \mathcal{P}\psi(2, 1) = \psi(1, 2) \tag{3.101}$$

so that

$$\mathcal{P}^2 = 1 \tag{3.102}$$

Taking ψ in particular to be an eigenfunction, ϕ, of $\mathcal{P}$ we have $\mathcal{P}^2\phi = \phi$ from Eq. (3.101) and further, from $\mathcal{P}\phi = \lambda\phi$, $\mathcal{P}^2\phi = \lambda^2\phi$. Hence $\lambda^2 = 1$. Thus the eigenvalues of $\mathcal{P}$ are $+1$ and -1; denoting the types of eigenfunction belonging to these eigenvalues by ϕ_+ and ϕ_- respectively, we have

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$$\mathcal{P}\phi_+(1, 2) = \phi_+(1, 2); \quad \mathcal{P}\phi_-(1, 2) = -\phi_-(1, 2) \tag{3.103}$$

In view of the definition of $\mathcal{P}$, these lead to

$$\phi_+(2, 1) = \phi_+(1, 2); \quad \phi_-(2, 1) = -\phi_-(1, 2) \tag{3.104}$$

so that ϕ_+ *and* ϕ_- *are symmetric and antisymmetric respectively under interchange of the particles.*

Now, applying $\mathcal{P}$ to $H(1, 2)\,\psi(1, 2)$ for arbitrary ψ, we obtain

$$\begin{aligned}\mathcal{P}H(1, 2)\,\psi(1, 2) = H(2, 1)\,\psi(2, 1) &= H(1, 2)\,\psi(2, 1) \\ &= H(1, 2)\,\mathcal{P}\psi(1, 2)\end{aligned} \tag{3.105}$$

where we have used the identity of particles, Eq. (3.99), in the penultimate step. Since Eq. (3.105) holds for arbitrary ψ, we have

$$\mathcal{P}H(1, 2) = H(1, 2)\,\mathcal{P} \tag{3.106}$$

Thus $\mathcal{P}$ commutes with $H(1, 2)$ and hence by the results of Sec. 3.14, represents a conserved quantity. Its eigenvalue is a good quantum number, i.e. if the wave function of the system at one instant is even (or odd) under interchange of particles, in the sense of Eq. (3.104), it will remain even (or odd) for all time. This can be seen directly from the Schrödinger equation $i\hbar\,\partial\psi(1, 2)/\partial t = H\psi(1, 2)$, which shows that $\partial\psi/\partial t$ is even or odd according as ψ is even or odd (since H itself is even). By repeatedly differentiating the equation with respect to time we see that $\partial^2\psi/\partial t^2$ and all higher derivatives have the same property. Hence the evenness or oddness of ψ is preserved always.

These considerations can be directly extended to a system of N identical particles ($N > 2$) by defining an exchange operator $\mathcal{P}_{\alpha\beta}$ for each pair of particles (α, β). What has been proved above in regard to $\mathcal{P}$ holds also for each $\mathcal{P}_{\alpha\beta}$. In particular, $\mathcal{P}_{\alpha\beta}\,H(1, 2, \ldots,\ldots, \alpha, \ldots, \beta, \ldots, N) = H(1, 2, \ldots, \alpha, \ldots, \beta, \ldots, N)\,\mathcal{P}_{\alpha\beta}$. Since $\mathcal{P}_{\alpha\beta}$ and H thus commute, any eigenfunction ϕ of H belonging to a non-degenerate eigenvalue must, according to Sec. 3.13, be also an eigenfunction of $\mathcal{P}_{\alpha\beta}$, i.e. $\phi(1, 2, \ldots, \alpha, \ldots, \beta,\ldots, N) = \mathcal{P}_{\alpha\beta}\,\phi(1, 2, \ldots, \beta, \ldots, \alpha,\ldots, N) = \lambda\phi(1, 2, \ldots, \beta, \ldots, \alpha, \ldots, N)$, $\lambda = \pm 1$. We have written the eigenvalue simply as λ instead of $\lambda_{\alpha\beta}$ because it is actually independent of (α, β). This fact is easily deduced from the observation that $\mathcal{P}_{\gamma\beta}\,\mathcal{P}_{\alpha\gamma}\,\mathcal{P}_{\alpha\beta} = \mathcal{P}_{\alpha\gamma}$: The action of the operators on the two sides of this equation on any N-particle wave function leads to identical wave functions. Operating with both sides of this operator equation on ϕ, one gets $\lambda_{\gamma\beta}\,\lambda_{\alpha\gamma}\,\lambda_{\alpha\beta} = \lambda_{\alpha\gamma}$, so that $\lambda_{\gamma\beta}\,\lambda_{\alpha\beta} = 1$. Hence $\lambda_{\gamma\beta}$ and $\lambda_{\alpha\beta}$ must *both* be $+1$ or *both* -1 (for arbitrary α, γ) showing that $\gamma_{\alpha\beta}$ is independent of α. The reader will have no difficulty convincing himself that it is independent of β too. We conclude thus that the wave function ϕ

must belong to one of two types: the *totally symmetric* (even) type which remains unchanged ($\lambda = +1$) under any interchange of particles, or the *totally antisymmetric* (odd) type which changes sign ($\lambda = -1$) with the interchange of any pair of particles.[11]

We now observe that wave functions of different symmetry types *cannot be connected* by any observable A of a system of identical particles, i.e. if ϕ is a totally symmetric/antisymmetric wavefunction, $A\phi$ is also totally symmetric/antisymmetric respectively. (This is because any observable must necessarily be a symmetric function of the variables of the particles which are indistinguishable.) Thus all the states of a given system, which are connected to one another by some observable or other, must belong to the same symmetry type. Since, as we have already seen, oddness or evenness does not change with time, it remains a permanent characteristic of the system. This property of quantum systems has no classical analogue.

It is a fact of observation that elementary particles found in nature do fall into two classes– those which always have totally antisymmetric wave functions, and those which have to be described by totally symmetric wave functions. We show below that 'particles which belong to the first category obey Fermi-Dirac statistics (and are hence referred to as *fermions*) while the others obey Bose-Einstein statistics (and are called *bosons*). Examples of fermions are electrons, protons and neutrons – all of which have a half-unit (½$\hbar$) of spin angular momentum – and any other particles with a half-integral value ($\frac{1}{2}$, $\frac{3}{2}$,...) of spin. When all the dynamical variables *including the spin variable* of two electrons (or two protons…) are interchanged, the wave function of the whole system (which may include other kinds of particles also) changes sign. Particles which have no spin (e.g. pi mesons) or have an integer value of spin angular momentum (in units of $\hbar$) are found to be bosons. Photons or light quanta (insofar as they can be regarded as particles) belong to this category. Interchange of any two identical particles of this kind leaves the wave function of the whole system unchanged. The fact that half-integral-spin particles are fermions and integral spin particles are bosons is the so-called *spin-statistics connection*, for which there are fundamental theoretical reasons. But these are beyond the scope of this book.

(*b*) *Relation between type of symmetry and statistics; The exclusion principle:* We have seen above that if a wave function ψ (1, 2) is to describe two identical particles it should not only satisfy the general admissibility conditions of Sec. 2.8,

[11] The arguments leading to this conclusion are clearly unaffected even if other particles (besides the identical particles on which we are focusing attention) are present in the system. The symmetry property of the wave function exists, however only with respect to interchanges of identical particles.

It is theoretically possible to conceive of identical particle systems which possess no nondegenerate energy levels and no complete set of commuting observables. [A.M.L. Messiah and O.W. Greenberg, *Phys. Rev.*, **136** *B*, 248 (1964)]. Such systems would have sets of wave functions which go over into combinations of themselves under interchange of particles, no single wave function being totally symmetric or antisymmetric. Their description cannot be done in terms of the ordinary Bose or Fermi statistics but requires the use of "parastatistics".

but also the further condition $\psi(1, 2) = \pm\psi(2, 1)$ where the sign depends on the kind of particle considered. Given some wave function $\phi(1, 2)$, we can construct ϕ_A or ϕ_S having the above property by symmetrizing or antisymmetrizing ϕ. Thus $\phi_A(1, 2) \equiv \phi(1, 2) - \phi(2, 1) = -\phi_A(2, 1)$ and $\phi_S(1, 2) \equiv \phi(1, 2) + \phi(2, 1) = +\phi_S(2,1)$.

Consider now the particular case when $\phi(1, 2)$ is the product of two single particle wave functions, say u_a and u_b, so that $\phi(1, 2) = u_a(1)\, u_b(2)$. This state can be described by saying that particle 1 has the wave function u_a and particle 2 has u_b. Such a description is not possible however if the particles are identical. If for example, we consider a system of two electrons, whose wave functions must be necessarily antisymmetric, we should use instead of the above ϕ,

$$\phi_A = u_a(1)\, u_b(2) - u_a(2)\, u_b(1) \tag{3.107}$$

The first term shows particle 1 as having the wave function u_a, while in the second term it is particle 2 which occupies u_a. All we can say then is that *one* of the particles occupies u_a and *the other* occupies u_b. The wave function does not permit us to specify *which* particle is where; the two particles are *indistinguishable.*

More generally, if we have a system of N electrons occupying single particle states described by wave functions $u_a, u_b, \ldots, u_g$ the state of the system as a whole must be described, not by a simple product wave function $\phi(1, 2, \ldots, N) = u_a(1)\, u_b(2) \ldots u_g(N)$ but by its antisymmetrized form which can be conveniently written as a determinant.[12]

$$\phi_A(1, 2, \ldots, N) = \begin{vmatrix} u_a(1) & u_a(2) & \ldots & u_a(N) \\ u_b(1) & u_b(2) & \ldots & u_b(N) \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ u_g(1) & u_g(2) & \ldots & u_g(N) \end{vmatrix} \tag{3.108}$$

Once again, it is possible to say only that one particle occupies u_a (but not *which* one), one particle occupies u_b, ...

One very important fact may now be noted. If $u_a = u_b$ in Eq. (3.107) or more generally, if any two or more of the functions $u_a, u_b, \ldots, u_g$ in Eq. (3.108) coincide, the wave function ϕ_A *of the system* vanishes – meaning that no such state exists. Thus *no two electrons can have the same wave function.* This is the famous *exclusion principle* first postulated by Pauli for explaining the periodic table of chemical elements. This principle is now seen to hold not only for electrons but for any system of particles having antisymmetric wave functions. Stating it slightly differently, *a given single particle state* (characterised by a particular wave function, say u_a) *cannot be occupied by more than one such particle.* It is a standard result in statistical mechanics that particles having this property must obey Fermi-Dirac statistics; they are fermions.

[12] This wave function is unnormalized. If $u_a, u_b, \ldots$ are members of an orthonormal set, the $N!$ terms arising from the expansion of the determinant may be seen to be separately normalized and orthogonal to each other. Then we must multiply (3.108) by $(N!)^{-1/2}$ to normalize it.

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For particles which are characterized by symmetric wave functions, there is no exclusion principle. For example $u_a(1)\, u_a(2) \ldots\, u_a(N)$ is a perfectly good symmetric function corresponding to all N particles having the same wave function u_a. But here also it is impossible to associate a specific particle with a particular u_a or u_b etc., as may be seen from the simple example of the two-particle wave function $\phi_S = u_a(1)\, u_b(2) + u_a(2)\, u_b(1)$. All one can say in general is that n_a particles occupy u_a, n_b occupy $u_b, \ldots$ This is precisely the kind of situation which leads to Bose-Einstein statistics in statistical mechanics. Thus particles described by symmetric wave functions are bosons.

It may be noted that tightly bound aggregates of particles (such as atomic nuclei) may themselves be viewed as 'particles' for the purpose of the above discussion, provided that circumstances are such that their internal structure does not come into play. Interchange of two *identical* nuclei, each made up of A fermions (neutrons + protons) is then equivalent to A simultaneous interchanges of *corresponding* pairs of particles (a proton of one nucleus with a proton of the other, etc.). Since each fermion interchange causes a change of sign, the overall effect on the wave function is given by a factor $(-1)^A$. Thus nuclei of even atomic number (e.g. He^4, C^{12}) behave as bosons while those of odd atomic number (e.g. He^3) behave as fermions.

The requirement of evenness or oddness of wave functions under interchanges of identical particles has important consequences, which we shall point out in ·the relevant context (theory of atomic and molecular structure, collision theory, etc.).

PROBLEMS

1. Compute $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p_x \rangle$, $\langle p_x{}^2 \rangle$ for a particle in one of the energy eigenstates in a square well potential. Verify that $(\Delta x)\,(\Delta p_x) \geqslant \frac{1}{2}\hbar$, where $(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$, $(\Delta p_x)^2 = \langle p_x{}^2 \rangle - \langle p_x \rangle^2$.
2. Verify explicitly that the different energy eigenfunctions of a particle in a square well potential are orthogonal.
3. Prove the quantum mechanical version of the virial theorem by showing that if T and V are the kinetic and potential energy operators, then $2\langle T \rangle = \langle \mathbf{x}.\nabla V \rangle$, and hence that if $V(r) = ar^n$, $2\langle T \rangle = n\langle V \rangle$.
4. If $u_1(\mathbf{x})$, $u_2(\mathbf{x})$ are degenerate eigenfunctions of the Hamiltonian $H = \mathbf{p}^2/2m + V(\mathbf{x})$, show that $\int u_1{}^*(\mathbf{x})\,(xp_x + p_x x)\, u_2\,(\mathbf{x})\, d\tau = 0$.
5. The wave function of a particle (in 3 dimensions) is $(x^2 + y^2)^{1/2} e^{-\alpha r}$. Normalize this wave function and calculate the expectation value of L_z.
6. Prove that for a particle in a potential V, $d\langle \mathbf{L} \rangle/dt = \langle \mathbf{T} \rangle$ where $\mathbf{T} \equiv -\mathbf{r} \times \nabla V$ is the torque. Note that this vanishes if V is a central potential.
7. Verify that for a particle in a potential, $\langle H \rangle$ can be written as $\int W\, d\tau =$ where $W = (\hbar^2/2m)\, \nabla\psi^*.\nabla\psi + \psi^* V \psi$. Show that W satisfies the continuity equation $\partial W/\partial t + \text{div}\, \mathbf{F} = 0$, where $\mathbf{F}$ is the energy flux vector. Determine the form of $\mathbf{F}$.

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8. Using the identity $[A,\, BB^{-1}] = 0$, show that $[A,\, B^{-1}] = -B^{-1}\,[A,\, B]B^{-1}$.

9. If $A,\ B$ are operators such that $[A,\, B] = 1$, and if $Z = \alpha A + \beta B$ where $\alpha,\ \beta$ are numbers, verify that $[A,\, Z^n] = n\beta Z^{n-1}$. Hence show that $(\partial/\partial\alpha)Z^n = nAZ^{n-1} - \frac{1}{2}\,n(n-1)\,\beta Z^{n-2}$ and $(\partial/\partial\alpha)e^Z = (A - \frac{1}{2}\beta\,)e^Z$. By integrating this differential equation, with due attention to the order of operator factors, show that .

$$e^{\alpha A\,+\,\beta B} = e^{\alpha A}\,e^{\beta B}\,e^{-\frac{1}{2}\,\alpha\beta}$$

10. If X and Y are any two operators, and $Z = e^{\alpha X}\,Y\,e^{-\alpha X}$, show that $dZ/d\alpha = [X,\, Z]$. Hence show that the Taylor expansion of Z (in powers of α) is

$$e^{\alpha X} Y\, e^{-\alpha X} = Y + \alpha\,[X, Y] + \frac{\alpha^2}{2!}[X, [X, Y]] + \frac{\alpha^3}{3!}[X, [X, [X, Y]] + \ldots$$

Show how the result of Prob. 9 can be obtained from this.